Still Echoing: Welfare Reform and the Noncognitive Skill Formation of the Next Generation

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Abstract

Noncognitive skills (NCS) are consequential dimensions of human capital and are shaped by early-life environments. While early-childhood interventions can have lasting effects on these skills, little is known about the long-run effects of social safety-net reforms on NCS. I examine the long-run effects of early-life exposure to the 1990s U.S. welfare reforms. Linking the Panel Study of Income Dynamics core survey to the Child Development Supplement and Transition into Adulthood Supplement, I exploit variation in reform exposure across states, birth cohorts, and mothers' likelihood of being affected in a triple-difference design. NCS are measured in young adulthood, when personality traits are more stable and developmentally consolidated. I find that early-life exposure reduced young-adult NCS by 0.23 standard deviations among individuals born to likely affected mothers. The decline is concentrated in agreeableness and openness and is largest among children exposed earliest in life. Effects are most detectable among girls and White children, although subgroup differences are imprecisely estimated. Suggestive mechanism evidence points less to household income or maternal employment and more to changes in early caregiving conditions. The findings show that safety-net reforms can leave lasting developmental imprints on the next generation.

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"Skills beget skills."
— James J. Heckman

1 Introduction

In the 1990s, the United States fundamentally restructured cash assistance for low-income families with children. Through state welfare waivers and the 1996 Personal Responsibility and Work Opportunity Reconciliation Act (PRWORA), Aid to Families with Dependent Children (AFDC), a federal entitlement, was replaced by Temporary Assistance for Needy Families (TANF), a block grant program built around work requirements, time limits, sanctions, and expanded state discretion. The reform sought to reduce welfare dependence, increase employment, and encourage family formation among the primarily single-mother households served by AFDC (Ziliak, 2015). In doing so, it changed not only the rules governing benefit receipt, but also the economic and caregiving environments in which low-income children were raised. Increased maternal employment could raise earnings and strengthen labor-market attachment, while sanctions, time limits, and unstable work could increase hardship, reduce parental time, and heighten household stress. The implications for child development were therefore ambiguous ex ante.

A substantial literature has sought to resolve this ambiguity by tracing welfare reform's consequences for families and children. Studies document effects on employment, welfare participation, income, family formation, and material hardship, as well as contemporaneous outcomes among children and adolescents, including cognitive development, schooling, health, behavior, delinquency, and risky behavior (Blank, 2002; Grogger, 2003; Ziliak, 2015; Bastian et al., 2021; Herbst, 2017; Kalil et al., 2023). More recent work has begun to follow exposed children into adulthood, examining educational attainment, food insecurity, family formation, and related outcomes (Corman et al., 2022; Dore et al., 2025; Vaughn, 2023). Yet we know considerably less about the reform's long-run human capital legacy for children exposed during the earliest years of life. This distinction matters because studies of young children generally observe outcomes while children are still young, while studies reaching adulthood often examine cohorts who were adolescents when reform occurred. Nearly three decades after PRWORA, children who experienced the reformed welfare regime during their earliest years have reached young adulthood, making it possible to ask whether those early exposures left a persistent developmental imprint.

I examine this question through noncognitive skills (NCS), an important but under-explored dimension of human capital in the welfare reform literature. NCS capture socioemotional traits and behavioral dispositions, including agreeableness, conscientiousness, emotional stability, and openness, that independently predict educational attainment, economic mobility, and health (Esping-Andersen and Cimentada, 2018; Heckman and Mosso, 2014; Humphries and Kosse, 2017). Their economic relevance may be growing as technological change increases the relative value of social and interpersonal skills in the labor market (Deming, 2017). Crucially, these skills remain malleable and responsive to parenting, stimulation, and caregiving environments throughout childhood and adolescence (Kautz et al., 2014), making them particularly relevant to a reform that altered the conditions under which low-income children developed. Nor can effects on NCS be inferred from evidence on cognitive skills, since the two dimensions follow different developmental trajectories and respond differently to environmental inputs. This distinction is consequential for how human capital is evaluated. As Heckman (2000) emphasizes, focusing on cognitive achievement alone can obscure the role of social adaptability, motivation, and other noncognitive capacities in shaping later outcomes. By measuring NCS in young adulthood, when personality traits have become more stable (Bleidorn et al., 2022), I ask whether welfare reform left a lasting imprint on this less visible dimension of children's human capital.

To answer this question, I use the Panel Study of Income Dynamics (PSID), linking the Child Development Supplement (CDS) to the Transition into Adulthood Supplement (TAS). These data follow children observed during welfare reform period into young adulthood and allow me to combine information on childhood family structure, maternal education, state of residence, and adult socioemotional outcomes. I measure exposure to welfare reform from conception through age five, the period in which changes in household resources, parental time, and caregiving routines are likely to be especially consequential for later skill formation (Cunha and Heckman, 2007). The empirical strategy exploits variation in the timing and intensity of AFDC waivers and TANF implementation across states, differences in exposure across birth cohorts, and differences in the likelihood of being affected by the reform based on family background. Specifically, I estimate a continuous-dose triple-difference design comparing children born to low-educated single mothers, the group most likely to be directly affected by welfare reform, to children from less affected family backgrounds, across states and birth cohorts.

I find that early-life exposure to welfare reform lowered NCS in young adulthood by about 0.23 standard deviations (SD) among individuals born to mothers most likely to be affected by the reform. The decline is concentrated in agreeableness and openness, with little evidence of changes in conscientiousness, extraversion, or neuroticism. The age pattern is consistent with a developmental interpretation, with the largest effects arising from exposure in the earliest years of life and smaller effects at older ages. Mechanism results suggest that welfare reform operated less through household income or maternal employment than through changes in early caregiving conditions. The effects are most visible among girls and White children, although the differences across groups are not precisely estimated. I then examine whether these socioemotional changes are reflected in adult economic outcomes. In descriptive Mincer-style regressions, agreeableness and openness are negatively associated with earnings, while conscientiousness and emotional stability are positively associated. Since the reform reduced traits that are not rewarded in the labor market, a simple decomposition does not predict an earnings loss and may even imply a gain. Consistent with this, I find no detectable effects on the same cohort's earnings or employment. The results suggest that welfare reform left persistent marks on socioemotional development, but along margins that standard economic outcomes may fail to capture.

These findings contribute to three strands of the literature. First, the paper extends the literature on the second-generation effects of welfare reform. Existing studies show that childhood exposure to welfare reform affected later outcomes such as educational attainment, fertility, family formation, and other measures of adult socioeconomic status (Bastian et al., 2021; Dave et al., 2012; Hartley et al., 2022; Herbst, 2017; Vaughn, 2022). I add to this literature by studying NCS in young adulthood, a dimension of human capital that has received limited attention in evaluations of welfare reform. The results show that the reform's long-run effects on the next generation were not confined to conventional outcomes such as schooling, employment, or family structure, but also extended to socioemotional traits that shape how individuals regulate behavior, interact with others, and navigate adult roles.

Second, the paper contributes to the broader literature on early-life environments and human capital formation. A central implication of skill-formation models is that early investments and family conditions can have lasting effects because skills are self-productive and because early capabilities affect the productivity of later investments (Cunha and Heckman, 2007; Heckman and Mosso, 2014). Consistent with this framework, I show that policy-induced changes in children's early household environments can be reflected in noncognitive skills measured many years later. The results therefore provide evidence that early-life exposure to public policy can leave persistent marks on socioemotional development, complementing evidence on the long-run effects of early childhood conditions on health, education, and adult socioeconomic outcomes (Bennhoff et al., 2024; Currie and Almond, 2011; García et al., 2023; Walker et al., 2022).

Third, the paper informs the literature on safety-net design and child development. Prior work on safety-net programs and long-run noncognitive outcomes has focused largely on unconditional cash transfers or in-kind supports (Akee et al., 2018; Baker et al., 2019; Bolbocean and Tylavsky, 2021; Milligan and Stabile, 2011). Welfare reform provides a distinct policy environment because it combined income support with work requirements, time limits, sanctions, and expanded state discretion. The findings show that the design of safety-net programs, not only the level of transfers, can shape the developmental trajectories of children in low-income families. This distinction is important for policy evaluation because reforms that alter both household resources and behavioral incentives may generate effects that are not fully captured by short-run measures of income, employment, or program participation.

The remainder of the paper proceeds as follows. Section 2 reviews the empirical evidence on welfare reform and children's human capital development. Section 3 presents the conceptual framework. Section 4 describes the data, exposure measure, and empirical strategy. Section 5 reports the main results. Section 6 discusses the findings and concludes.

2 Empirical Evidence

Whilst the impact of the reform on the immediate beneficiaries has been well established, documenting large and well-identified impacts on maternal employment, welfare caseloads, and household income (Blank, 2002, 2009; Grogger and Karoly, 2005; Ziliak, 2015). The impact on the second generation has received limited attention, particularly the human capital development of the children of the affected mothers. Here, I stratify the existing evidence on it's human capital impact into cognitive and noncognitive.

The foundational insight of the cognitive strand comes from experimental evidence. Morris et al. (2009), synthesizing seven welfare experiments conducted across the United States in the 1990s, find that programs simultaneously boosting maternal employment and raising household income through earnings supplements improved young children's achievement scores by approximately 7 percent of a SD, with effects concentrated among children aged five and under. Programs mandating work without supplementing earnings produced no corresponding cognitive gains, establishing that material stability rather than maternal employment per se is the operative developmental input. Quasi-experimental work using variation in state-level reform timing confirmed and extended these signals. Welfare reform improved test score performance among low-income students, with gains concentrated in states experiencing the greatest caseload reductions (Miller and Zhang, 2009), and structural simulation evidence reinforces the mechanism, combining time limits with work requirements increased cognitive attainment by altering maternal employment and welfare use decisions (Chyi et al., 2014; Chyi and Ozturk, 2013). These cognitive improvements carried forward into longer-run human capital accumulation, manifesting in higher enrollment rates, lower dropout rates, and greater college completion among exposed children (Dave et al., 2012; Offner, 2005; Vaughn, 2022). Where adverse effects emerge they trace the same mechanism in reverse, welfare receipt is associated with cognitive score declines operating through maternal stress and income pathways under unstable employment (Heflin and Acevedo, 2011; Herbst, 2017), and attainment effects vary by gender and policy bundle (Bastian et al., 2021).

Across this strand, the evidence is coherent and conditional, reform moved cognitive and educational trajectories when and because it altered household material conditions. What it cannot tell us is whether the reform shaped the formation of socioemotional traits that are distinct from cognitive ability, differently determined, and independently consequential for long-run human capital.

Complementary strand of the literature motivated by the same query on how the welfare reform altered the household conditions of early childhood, those alterations should have registered not only in test scores but in how the socioemotional and behavioral development of the affected children directly and indirectly. In similar fashion to the cognitive literature, experimental evidence on programs incorporating welfare reform features produced heterogeneous effects on externalizing behaviors, with improvements in some contexts and adverse effects in others, depending critically on whether earnings supplements accompanied the work mandate (Chase-Lansdale et al., 2011; Duncan et al., 2007; Gennetian and Miller, 2002; Gennetian et al., 2004). Chase-Lansdale et al. (2011) find that youth whose mothers increased employment net of welfare participation were less likely to exhibit serious behavior problems, reinforcing the interpretation that income stability, not employment itself, drives behavioral outcomes. Other studies sharpened this mechanism by organizing behavioral outcomes around the nature of the maternal transition rather than the fact of employment. Where transitions off welfare were abrupt or precarious, children exhibited elevated aggressive and withdrawn behavior (Hofferth, 1999).

In contrast, where transitions were gradual and combined welfare receipt with work, both internalizing and externalizing problems declined (Dunifon et al., 2003), and stable maternal employment during early childhood predicted further reductions in externalizing behaviors without any corresponding cognitive gains (Pilkauskas et al., 2018). Aughinbaugh and Gittleman (2004) reinforce the conditionality of these effects by finding no impact of maternal employment on risky behaviors, confirming that work alone, absent the stability and income gains that make employment consequential, leaves behavioral trajectories unmoved. In a parallel study, Herbst (2017) exploiting the variation in AYCE lever of the welfare reform observed no impact on affected children's socioemotional development proxied by their externalizing behavior, prosocial behavior and approaches to learning.

Among adolescents, studies examining risky and antisocial behaviors, outcomes closely related to NCS, also yield mixed findings. Exploiting variation in the implementation of the welfare reform across states and over time, Corman et al. (2017a) finds that adolescent aged 15-17 experienced a reduction in minor crimes by 9-11 percent, however, using similar identification strategy in a companion paper they find no effect of the reform on drug related arrests Corman et al. (2017b), but they do find some modest increase in arrest. Shifting from high-stake outcomes, Dave et al. (2021), also found that the welfare reform rather led to an increase in engagement of risky behavior among adolescents, with the effect being more pronounced for boys as against girls. Recent work by Reichman et al. (2023), extending the outcome set to positive health and social behaviors, find no robust effects on any prosocial dimension. This set of literature also shows how the welfare reform affected the behavioral and socioemotional component of the children and adolescents of the welfare families, thereby directly and indirectly show the noncognitive impact of the reform. However, these findings are largely contemporaneous, capturing the immediate reactions to the reform rather than the enduring NCS that crystallize across early childhood and persist into adulthood.

Across both strands, the literature converges on a finding that is as consistent as it is incomplete. The impact of the welfare reform had substantial impact the human capital of the affected children and adolescents, but the effect was heterogeneous and in every case conditioned by the quality and stability of the household environment the reform created. When reform improved material conditions, cognitive scores rose and behavioral problems fell. When it destabilized them, the reverse obtained. This pattern is not a contradiction, it is the same mechanism operating across different outcome domains, pointing consistently toward the early-life household environment as the operative channel through which welfare reform reached the next generation.

What that convergent finding cannot resolve is whether welfare reform altered the generative layer through which all of these observable outcomes are produced. Both strands measured what children achieved or exhibited rather than the socioemotional traits that produced those outcomes. Critically, much of the existing evidence is contemporaneous, capturing children's outcomes in the years surrounding the reform rather than the socioemotional traits that become more stable and consequential as exposed cohorts enter early adulthood. To my knowledge no study has exploited a validated psychometric measures of NCS to assess how the reform conditioned socioemotional development into young adulthood. This gap is consequential as evidence indicates that children raised in poverty and disadvantaged environments carry lasting NCS deficits into adulthood (Heckman, 2008; Phillips and Shonkoff, 2000). A reform that moved those inputs at scale during the critical early-life window did not merely shift test scores and behavioral responses; it may have recalibrated the socioemotional foundations from which those outcomes derive. That possibility has not been examined, and this paper provides the first direct evidence on it.

3 Theoretical Model

Understanding whether welfare reform shaped the noncognitive development of exposed children requires a framework that links policy-driven changes in early-life environments to the long-run accumulation of socioemotional skills. The relevant question is not whether reform altered parental behavior in the abstract, but whether it moved the specific inputs through which NCS formation proceeds during the developmental window when those inputs are most consequential. This section formalizes that link, derives the theoretical predictions the empirical design tests, and identifies the channels through which welfare reform plausibly altered the early-life environment.

3.1 Skill Formation Technology

I adopt the multistage skill formation model of Cunha and Heckman (2007). Let $t \in \{0, 1, 2\}$ denote prenatal and infancy ($t = 0$), early childhood ($t = 1$), and young adulthood ($t = 2$). Let σ_{it} represent individual i's NCS stock at stage t. Skills evolve through a recursive production technology:

$$\sigma_{i,t+1} = f^t(\sigma_{it}, I_{it}, B_i, \nu_{it}) \tag{1}$$

where B_i captures stable family background factors, I_{it} represents parental investments and home environments during stage t, and ν_{it} is an idiosyncratic shock. Two properties of this technology structure the empirical interpretation.

Self-productivity implies that early skills are persistent:

$$\frac{\partial f^t}{\partial \sigma_{it}} > 0 \tag{2}$$

so early socioemotional traits reinforce themselves across developmental stages. A child who enters early childhood with stronger self-regulation and emotional stability is better positioned to benefit from subsequent parental and institutional investments.

Dynamic complementarity implies that early skill stocks raise the productivity of later investments:

$$\frac{\partial^2 f^t}{\partial \sigma_{it}\, \partial I_{it}} > 0 \tag{3}$$

The stage-specific sensitivity embedded in this technology implies that the in-utero to age-five window is uniquely consequential. Neuroscientific and developmental evidence establishes that the prefrontal cortex, the neurological substrate of self-regulation and emotional control, undergoes its most rapid development during this period, making socioemotional traits particularly responsive to environmental inputs before age five and substantially less malleable thereafter (Cunha and Heckman, 2007; Kautz et al., 2014; Phillips and Shonkoff, 2000). Addressing deficits that emerge in this window at later stages is a poor remediation strategy: the complementarity structure implies that early shortfalls in σ_{i1} reduce the productivity of all subsequent investments, so the costs of environmental deprivation during early childhood compound rather than dissipate over the life course.

Iterating equation (1) yields a cumulative expression for young-adult NCS:

$$\sigma_{i2} = h(B_i, \sigma_{i0}, \bar{I}_{i0}, \bar{I}_{i1}) \tag{4}$$

which establishes that persistent differences in young-adult NCS reflect the joint history of early inputs and family background. Any policy that shifts I_{it} during the early childhood window therefore has the potential to alter the socioemotional trajectory from which all subsequent human capital accumulation derives.

With some assumptions, equation (4) can be taken to the data. I assume that the production function g is linear in inputs and constant across time and individuals. Young-adult NCS, σ_{i2}, are proxied by validated Big Five personality measures from the PSID TAS, which capture the stable socioemotional traits that crystallize by young adulthood (Bleidorn et al., 2022). Maternal education serves as a proxy for the initial skill endowment σ_{i0}, reflecting the family background conditions that shape the early developmental environment. Early childhood exposure to welfare reform, constructed as the share of the in-utero to age-five window spent under post-reform rules, proxies for family investment decisions I_{it}, capturing the policy-induced variation in early-life inputs that the triple-difference design exploits.

Welfare reform enters this framework by altering I_{it}, the early investments and home environments that govern NCS formation. Let $W_{s,t}$ denote the welfare regime in state s and year t. Children of low-educated single mothers, the group most directly exposed to AFDC waiver and TANF provisions (Bitler and Hoynes, 2010), experienced substantial shifts in both material and psychosocial conditions as states moved from unconditional cash assistance toward a work-conditioned, time-limited regime. These shifts operated through two intertwined channels, each with distinct sub-mechanisms.

The economic investment channel operates through household resources and material stability. Welfare reform increased labor supply among single mothers (Grogger and Karoly, 2005; Ziliak, 2015), but greater employment did not necessarily translate into higher or more stable household resources, particularly when affected mothers entered low-wage or unstable work or lost benefits as earnings rose. Where reform increased disposable income and reduced hardship, it could strengthen investments in children and support NCS formation (Schoeni and Blank, 2000; Akee et al., 2018; Fletcher and Wolfe, 2016; Morris et al., 2009). Where earnings gains were limited or offset by benefit losses and greater instability, the same channel could instead weaken the material environment in which children developed. Household composition may further amplify or dilute these effects by changing the resources available per child.

The caregiving-disruption channel operates through maternal time, stress, and the continuity and quality of care. Work requirements, sanctions, and time limits could reduce parental availability, increase caregiving demands, and shift children toward alternative care arrangements even without materially changing household income (Heflin and Acevedo, 2011; Kalil et al., 2023). These changes may be especially consequential during the in-utero-to-age-five window, when socioemotional development is particularly sensitive to stable and responsive caregiving. The sign of this channel remains ambiguous if employment also introduces greater routine or stability into family life.

These two channels can be summarized by a policy-to-investment mapping:

$$I_{it} = g^t(W_{s(i),t}, B_i, u_{it}) \tag{5}$$

where u_{it} captures unobserved household heterogeneity and $s(i)$ denotes child i's state of residence. Because $W_{s,t}$ shifted discretely across states and years as states adopted AFDC waivers and implemented TANF at different points in time, children born in different states and cohorts experienced different shares of their prenatal-to-age-five window under the reformed regime. This discrete cross-state and cross-cohort variation in $W_{s,t}$ generates the exposure index that serves as the identifying source of variation in the empirical design.

The framework generates two predictions. First, if welfare reform altered inputs to NCS formation during early childhood, greater exposure should affect young-adult NCS, with effects strongest for exposure earliest in the developmental window under the critical-period logic. Second, the sign is theoretically ambiguous because both the economic-investment and caregiving channels can improve or worsen the environments in which NCS develop. The net effect therefore depends on how reform changed household resources, material stability, and caregiving conditions.

Anchored in this framework, the triple-difference design estimates the effect of welfare-reform-induced variation in these early-life inputs on young-adult NCS. The estimates therefore capture whether restructuring cash assistance during a sensitive developmental period left a lasting socioemotional imprint on exposed children.

4 Data

This study leverages the PSID dataset, the world's longest-running nationally representative household panel, along with its CDS and TAS. The PSID began in 1968 with roughly 18,000 individuals in about 4,802 households and has since followed original sample members and their descendants annually through 1997 and biennially thereafter, collecting detailed socioeconomic, demographic, and geographic information. The CDS and TAS were designed to facilitate research on the intergenerational well-being of children and young adults within their family, school, and neighborhood contexts. The CDS, launched in 1997, sampled children born between 1984 and 1997 who were ages 0–12 at baseline. Up to two children per core family were randomly selected and interviewed, alongside their primary caregivers, and were followed until they reached age 18. Upon turning 18, these individuals transitioned into the TAS. The first TAS wave began in 2005 and continues to track these same respondents biennially through approximately age 28, gathering information on educational trajectories, labor-market experiences, social attitudes, and psychological and socio-emotional well-being. Thus, the PSID and CDS–TAS structure provides a rare longitudinal framework linking childhood environments to early-adult outcomes within the same individuals. The structure of the PSID-CDS-TAS sample is shown in Figure A1.

The PSID is particularly well suited for this study for four reasons. First, the CDS baseline coincides with the 1990s welfare reform era coupled the kids birth year and month, enabling credible identification of children developmentally exposed to the policy. Second, the PSID provides geocoded state identifiers, allowing linkage to AFDC waiver and TANF implementation timing and facilitating cross-state comparisons. Third, its intergenerational design yields rich information on both children and parents, supporting detailed controls for household socioeconomic conditions and parental behaviors. Fourth, TAS includes widely used and psychometrically validated measures of NCS, the primary outcomes of interest (Goldberg, 1993; Kautz et al., 2014). The PSID and its supplements have been extensively used to examine the intergenerational consequences of early-life policy exposure and environmental shocks (e.g., Dore et al., 2025; Corman et al., 2022 ; Hoynes et al., 2016; Vaughn, 2023). As with most long-running panels, the PSID experiences differential nonresponse and attrition, and oversamples low-income families; however, these features are mitigated using the survey weights provided. Finally, for the welfare reform timing, I extract the data from US Department of Health and Human Services and (Crouse, 1999). Childhood level state macroeconomic variables were also extracted from the University of Kentucky Center for Poverty Research Welfare Data (UKCPR, 2026).

4.1 Measures

4.2 Welfare Reform Exposure

Following the exposure-share approach of Hoynes et al. (2016), I construct a continuous measure of early-life exposure to welfare reform, defined as the fraction of months between in-utero and age five during which a child resided in a state operating either an AFDC waiver or TANF. I consider the in-utero period because prior scholarship has documented how welfare reform affected maternal prenatal decisions, investments, employment, and stress, shaping the early developmental environment (Kaestner and Chan Lee, 2005). I first assign each child's policy environment based on their birth state and the month-year in which that state first implemented an AFDC waiver or TANF. I convert these dates into monthly format and compute exposure relative to the child's birth month-year. For each cohort, I measure exposure over a 69-month window (9 months in utero plus 60 months after birth). Let B_s denote the child's birth state, t_b the birth month-year, and t_s the first reform month-year for state s. The exposure window spans $[t_b - 9,\ t_b + 60]$. This measure takes the value of 0 for children born entirely before any reform or in states that adopted only after the child aged out of the window, and 1 for children fully exposed from conception to age five. For children whose exposure falls between fully treated and untreated, I quantify dosage as the proportion of the first 69 months of life, from in-utero to age five, spent in a state operating under a major waiver or TANF,

$$\text{Exposure}_i = \frac{\text{Months exposed in state } B_s}{69}.$$

Because all reforms occurred prior to data collection, this exposure is treated as time-invariant. This construction improves upon simple contemporaneous reform indicators by capturing the intensity and timing of early-life policy exposure rather than relying solely on residence during a single survey year. Consistent with prior work (Bitler and Hoynes, 2010), welfare reform is expected to primarily affect children of single mothers with a high school education or less. I therefore define this group as the policy-relevant (target) population. The comparison group includes children of single mothers with more than a high school education and those raised in two-parent households.

4.3 Noncognitive Skills

I measure NCS using validated items from the PSID-TAS data spanning 2013 to 2023, which capture the five Big Five personality domains, namely Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism (Goldberg, 1993; Kautz et al., 2014). Openness reflects curiosity and receptiveness to new ideas; Conscientiousness captures diligence and reliability; Extraversion reflects sociability and external engagement; Agreeableness gauges prosocial orientation; and Neuroticism captures emotional instability and susceptibility to stress. The survey items corresponding to each domain are listed in Table A4. To construct a composite NCS index, I follow the approach of Kling et al. (2007). First, all items are recoded so that higher values consistently reflect more desirable traits. Next, I compute item-level z-scores by subtracting the comparison-group mean and dividing by the comparison-group standard deviation, expressing each trait in standard-deviation units relative to the untreated distribution. Finally, I take the mean of all available standardized items for each respondent. This aggregation improves reliability, reduces measurement error, and avoids the arbitrary weighting and multiple-testing concerns inherent in domain-specific analyses. I then replicate the same procedure for each subscale.

4.4 Covariates

The analysis includes a rich set of controls to account for demographic composition, family background, and state-level policy environments relevant to children's early-life conditions. At the individual level, I adjust for sex, age in adulthood, race (Black and other non-White groups, with White as the reference category), and an indicator for being raised by a grandparent to capture alternative caregiving arrangements. Maternal education and maternal age are included as core measures of socioeconomic status and parental human capital. To account for pre-reform economic and policy conditions in the child's state, and to mitigate bias arising from nonrandom post-reform migration, I incorporate a set of pre-reform state characteristics, namely the unemployment rate, log TANF and log SNAP benefit levels, the poverty rate, the minimum wage, and the generosity of the Earned Income Tax Credit (EITC), each interacted with birth year to allow their influence to vary across cohorts. These factors jointly capture variation in state economic opportunities and safety-net generosity during the birth-cohort window, helping isolate the effect of early-life welfare reform exposure from broader policy and labor-market environments.

5 Empirical Strategy

I estimate the long-run effect of early-life exposure to welfare reform on NCS using a difference-in-difference-in-differences (DDD) design. The design exploits three sources of variation: the staggered timing of reform across states, differences in exposure across birth cohorts within states, and differences in families' likelihood of being affected by the reform. Children born to low-educated single mothers form the target group because these families were most likely to be exposed to AFDC and TANF incentives, while children of more-educated single mothers and those in two-parent households form the comparison group (Bitler and Hoynes, 2010; Dave et al., 2021). The DDD therefore asks whether greater early-life exposure differentially changed NCS among children whose families were most likely to be affected, relative to children facing the same state and cohort environments but substantially less exposure to the reform. The baseline specification is as follows:

$$NCS_{isct} = \delta_1 RE_{isc}^{\text{IU-5}} + \delta_2 \left(RE_{isc}^{\text{IU-5}} \times T_i\right) + \beta X_{isct} + \alpha_s + \gamma_c + \eta_t + \theta_T + \varepsilon_{isct}, \tag{5}$$

where i denotes the individual, s the birth state, c the birth cohort, and t the survey year. $RE_{isc}^{\text{IU-5}}$ measures the share of the in-utero-to-age-five window spent under welfare reform, so treatment is a continuous exposure dose rather than a binary indicator. T_i identifies the target group. Because target status is time invariant, its level difference is absorbed by the target-group fixed effect, θ_T. X_{isct} contains child, maternal, and state-level controls, while α_s, γ_c, and η_t denote birth-state, birth-cohort, and survey-year fixed effects.

The coefficient of interest is δ_2, which captures the differential effect of an additional share of the early-life window under reform for target children relative to the comparison group. The coefficient δ_1 captures the corresponding exposure gradient among comparison children. Because exposure is assigned by state and birth cohort rather than realized welfare participation, δ_2 is an intent-to-treat effect of exposure to the reformed welfare regime. I expect δ_1 to be small because comparison families were substantially less likely to be affected by welfare reform, making it a useful internal diagnostic of the comparison group.

A distinctive feature of welfare reform is that it was permanent rather than episodic. Once a state implemented an AFDC waiver or TANF, it did not return to the pre-reform regime. A child exposed during early childhood was therefore generally exposed later in childhood as well. The design consequently does not identify an ever-exposed versus never-exposed contrast. Instead, it identifies the effect of additional exposure during the in-utero-to-age-five window, conditional on exposure to the reformed regime later in childhood. This is the relevant estimand for the paper's focus on whether exposure during the earliest developmental years had lasting consequences for NCS.

The central identifying assumption is that, absent welfare reform, the difference in NCS between the target and comparison groups would have evolved similarly across birth cohorts within states. Because both groups share the same state-level reform timing and broader state and cohort environments, the DDD differences out shocks common to them. Identification would fail if an unobserved factor coinciding with reform timing differentially affected target families relative to comparison families in a way correlated with early-life exposure. This is the core counterfactual condition underlying the estimates.

Standard errors are clustered at the state level because reform timing and provisions vary at that level. Given the limited number of state clusters (36), I also report wild cluster bootstrap p-values based on 9,999 replications with Rademacher weights.[1]

5.1 Threats to Identification

The credibility of this design ultimately depends on whether alternative forces could generate the same target-comparison pattern around reform. I therefore subject the baseline estimates to a series of tests that address the main threats to identification, including differential cohort trends, endogenous reform timing, selective migration, and complications from staggered adoption.

5.1.1 Differential Trends

A first concern is that target and comparison children may have followed different developmental trajectories even in the absence of welfare reform. I address this concern at several levels. In the preferred specification, I interact pre-reform state economic and policy characteristics with birth cohort. These interactions allow states entering the reform period with different unemployment rates, poverty rates, AFDC and SNAP caseloads, minimum wages, and EITC generosity to follow different cohort trajectories rather than imposing a common relationship between these initial conditions and later NCS.

I then add state-specific linear birth-cohort trends, $\alpha_s \cdot c$, which allow each state to follow its own systematic cohort trajectory. This specification provides a stronger adjustment for smooth differences in trends across states, although I interpret it cautiously because flexible trends may absorb part of the treatment response along with potential confounding (Wolfers, 2006). For this reason, the specification without state-specific trends remains preferred.

Finally, I examine the parallel-trends implication directly using an event-study specification. I replace the continuous exposure measure with indicators for years relative to state reform adoption, each interacted with target status, and omit the year immediately preceding adoption as the reference:

$$NCS_{isct} = \sum_{k \neq -1} \pi_k \left(D^k_{sc} \times T_i \right) + \beta X_{isct} + \alpha_s + \gamma_c + \eta_t + \theta_T + \varepsilon_{isct}, \tag{6}$$

[1]Following Roodman et al. (2019), the wild cluster bootstrap provides more reliable inference when the number of clusters is limited and conventional cluster-robust inference may over-reject.

where D_{sc}^{k} indicates that cohort c in state s falls k years relative to reform adoption. The coefficients $\{\pi_k\}_{k<-1}$ trace the target-comparison gap before reform could affect the relevant cohorts. A lack of systematic pre-reform divergence supports the assumption that the two groups were not already moving along different trajectories before policy exposure. The post-reform coefficients also show when the target-comparison gap begins to emerge relative to reform implementation.

5.1.2 Endogenous Reform Timing

A second concern is that states may have adopted welfare reform in response to economic or policy conditions that independently shaped children's later development. If states experiencing unusually favorable or adverse conditions systematically reformed earlier, the exposure measure could partly capture those underlying conditions rather than welfare reform itself.

In addition to controlling flexibly for pre-reform state conditions, I examine whether reform timing is predictable from those conditions. I measure reform timing as the number of months from January 1992 to the earlier of a state's AFDC waiver or TANF implementation and regress this measure on pre-reform state characteristics. I estimate these relationships across the reform states and assess their sensitivity to Census-region fixed effects, population weighting, and the exclusion of influential states. Little systematic relationship between pre-reform conditions and adoption timing would support the interpretation that the cross-state timing variation used in the design is not primarily driven by observable state circumstances relevant to later NCS.

5.1.3 Selective Migration

A third concern is selective migration. Because exposure is assigned using state of birth, families that moved across states during the relevant childhood window may have experienced a different reform regime from the one assigned to them. Migration could therefore introduce exposure measurement error and, if relocation itself responded to welfare policy or correlated with unobserved family characteristics, could also bias the DDD estimate.

I assess this concern by re-estimating the baseline specification among children whose birth state matches their state of residence. For these children, the link between assigned state and observed residence is substantially cleaner. I build the specification from the fixed-effects model through the preferred control set and compare the resulting DDD estimates with those from the full sample. Stability of the coefficient provides evidence that cross-state migration is unlikely to account for the baseline result.

5.1.4 Staggered Adoption and Negative Weighting

A final concern arises from the staggered implementation of welfare reform. With heterogeneous treatment effects, fixed-effects estimators may place problematic or negative weights on some treatment comparisons and may use already-treated units as controls for later-treated units, including in settings with continuous treatment (de Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021; Callaway et al., 2024). My treatment measure differs from the conventional contemporaneous treatment indicator because it represents a completed early-life exposure dose fixed long before NCS is observed. Nevertheless, states adopted reform at different times, so the baseline estimator still combines comparisons across adoption cohorts whose effects may differ.

I assess this issue in two ways. First, I decompose the fixed-effects estimand into its underlying group-by-period treatment effects and examine the weights attached to those comparisons (de Chaisemartin and d'Haultfoeuille, 2020). This exercise establishes whether the baseline coefficient receives meaningful influence from negatively weighted comparisons.

Second, I estimate a stacked DDD that restricts comparisons to treated and not-yet-treated states. Because all states had implemented either an AFDC waiver or TANF by 1997, states reforming in 1997 cannot form a treatment stack with clean controls. I therefore construct treatment stacks for adoption cohorts through 1996 and use the same sample window in the corresponding conventional DDD estimate to maintain comparability.

Each adoption cohort c defines a sub-experiment in which treated children are those in states reforming in year c, while comparison states are those that had not yet implemented reform. After trimming the sub-experiments to a common event window and stacking them, I estimate

$$NCS_{isce} = \delta_1 RE_{isc}^{\text{IU-5}} + \delta_2 \left(RE_{isc}^{\text{IU-5}} \times T_i\right) + \beta X_{isce} + \alpha_{se} + \gamma_{ce} + \eta_{te} + \theta_{Te} + \varepsilon_{isce}, \tag{7}$$

where e indexes the adoption-specific sub-experiment. The estimating equation otherwise mirrors the baseline DDD, but the fixed effects are allowed to vary across sub-experiments. This construction confines identification to comparisons between treated and not-yet-treated states within each adoption cohort and removes comparisons in which already-treated states serve as controls. Agreement between the conventional and stacked estimates would therefore indicate that the baseline result is not driven by problematic comparisons created by staggered adoption.

6 Results

6.1 Descriptives

Table 1 reports survey-weighted means and SD for the analysis variables, and Table A2 compares the target and comparison groups. The average child spent roughly 21 of the 69 months in the in-utero to age-five window under reformed welfare rules, though exposure varies widely across cohorts and states, with a standard deviation of about 22 months spanning children with almost no exposure to those exposed across the entire window. About 15 percent of the sample falls in the target group, children of low-educated single mothers. As expected by construction, the groups differ most on the dimensions that define the target. Target children are far more likely to have a mother without a high school diploma, at 45 percent against 10 percent among the comparison group, and to be nonwhite, at 63 percent against 20 percent, both differences large in standardized terms. The pre-reform state characteristics are comparable across groups, with small, standardized differences for most and only the poverty rate showing a nontrivial gap, at 14.6 against 13.3. That target and comparison families faced broadly similar pre-reform state environments supports the identifying assumption that reform exposure is not standing in for divergent state economic conditions.

[Table 1 About Here]

Appendix Figure A4 plots kernel densities of NCS for the two groups across the composite index and the five subscales. The composite distributions are nearly coincident, while the subscales separate to differing degrees, with agreeableness and openness showing the clearest divergence. These are raw, unconditional comparisons that reflect group differences in background as much as any effect of reform, and the triple-difference design isolates the latter, to which I turn below.

Figures A2 and A3 document the rollout of reform. No state operated an AFDC waiver before 1992, and adoption rose sharply thereafter, so that by PRWORA's enactment in 1996 most states had an active reform, with all states covered by 1997. This staggered adoption generates the cross-state, cross-cohort variation in early-life exposure that identifies the estimator. Figure A2 shows that early adopters were geographically dispersed across the South, Midwest, and West rather than concentrated in a single region, mitigating the concern that the variation reflects regional economic trends rather than welfare policy.

6.2 Baseline Results

I present the main estimates in Tables 2 through 5 using the triple-difference design described in Section 4. Two features of the specification are important for interpretation. First, IU-Age 5 measures the share of the in-utero to age-five window that a child spent under welfare reform. Its interaction with target status therefore captures the effect of moving from no exposure to full exposure during early childhood for children born to mothers most likely to be affected by the reform. Because all NCS outcomes are standardized, the coefficients are interpreted in standard deviation units. Second, treatment is assigned at the state-by-birth-cohort level rather than by realized welfare participation. The estimates are therefore best interpreted as intent-to-treat effects of early-life exposure to the reformed welfare regime among children born to mothers most likely to be affected. They capture the effect of policy exposure, not the effect of actual welfare receipt.

[Table 2 About Here]

Table 2 develops the composite estimate across five specifications of increasing richness. Column 1 absorbs only the fixed effects; columns 2 through 4 add maternal characteristics, young adult controls, and pre-reform state economic conditions interacted with birth cohort; and column 5 adds state-specific linear birth-cohort trends. Consistent with my treatment of the trend specification in Section 4, I take column 4 as preferred, since the linear cohort trends in column 5 risk absorbing part of a dynamic treatment response. The estimate is remarkably insensitive to this sequence. IU-Age 5 x Target enters at −0.23 SD with fixed effects alone and holds at −0.23 SD through the full control set, moving only trivially once trends are added. That the coefficient is unmoved by the progressive inclusion of maternal, child, and state covariates is the first indication that it reflects the reform rather than the correlation of exposure with background. The coefficient on IU-Age 5 alone, which recovers exposure among the comparison group, is a precise zero throughout, as expected for children the policy was not designed to reach.Because state clusters are few, conventional standard errors could overstate precision, so I report wild cluster bootstrap p-values throughout; the DDD estimates retain significance under this stricter standard.

[Table 3 About Here]

Table 3 then decomposes the composite into its five subscales under the preferred specification, and the decline proves selective rather than uniform. It concentrates in agreeableness and openness, at −0.31 SD and −0.44 SD, both significant at five percent. The remaining traits, conscientiousness (−0.16 SD), extraversion (−0.07 SD), and neuroticism (−0.14 SD), are smaller and fall short of significance, though all five estimates share the same negative sign. Because the subscales form a family of five related hypotheses, I report Benjamini et al. (2006) sharpened two-stage q-values controlling the false discovery rate at 0.10. Agreeableness and openness each survive the correction at a q-value of 0.04; the remaining three do not and the remaining three do not, and these two subscales likewise retain significance under the wild cluster bootstrap.

[Table 4 About Here]

Table 4 estimates the specification separately by child sex. The effect is carried by girls, whose total effect is −0.26 SD on the composite at one percent and −0.37 SD on agreeableness at five percent, while the estimates for boys are muted and statistically indistinguishable from zero. This asymmetry warrants care. The triple interaction, IU-Age 5 x Target x Female, is itself insignificant, so although the effect is precisely estimated among girls, the data do not support the stronger claim that it exceeds the effect for boys.

I read the estimates as marking where the effect is identified rather than as a contrast between the sexes, and I hold to that distinction throughout the heterogeneity analysis.

[Table 5 About Here]

Table 5 repeats the exercise by child race, and the pattern parallels the one for sex. The declines concentrate among White children, at −0.29 SD on the composite, −0.47 SD on agreeableness, and −0.61 SD on openness, each significant at five percent or better, while the total effects for nonwhite children are smaller and never significant. As before, the triple interaction is insignificant, so I take these estimates to locate the subgroup in which the effect is most precisely estimated rather than to establish a genuine difference between White and nonwhite children.

6.3 Mechanisms

I examine the channels through which reform exposure operates using the 1997 Child Development Supplement. One caveat frames what follows. The supplement records household and caregiving conditions at a single point rather than continuously across the exposure window, so these variables capture a snapshot of the early environment around 1997 rather than the full trajectory from in-utero to age five. They are best read as indicators of which inputs the reform moved, not as a complete time-path of exposure. Table 6 reports triple-difference estimates for each candidate channel, spanning childcare arrangements, household income, maternal employment, maternal distress, the presence of a young child, and mother-child closeness.

[Table 6 About Here]

The estimates point more strongly to the caregiving environment than to household economic resources. Reform-exposed target children are 8 percentage points more likely to be in formal childcare and 11.4 percentage points more likely to live with another young child in the household, while informal care is unchanged. The economic margins tell a different story. Household income barely moves and is imprecisely estimated, and maternal employment is effectively zero. These estimates provide little evidence that the reform raised the material resources that might have buffered children. Maternal distress is positive but imprecise, and mother-child closeness is unchanged.

A feature of the policy lets me probe the caregiving reading more directly. If the harm works by displacing early maternal care, it should fade where the reform left that care intact, and states differed sharply in exactly this respect. The age-of-youngest-child exemption set how young a child could be before its mother faced work requirements, ranging from immediate mandates in the least generous states to multi-year shelters in the most generous. Table 7 lets the treatment effect vary with this threshold, drawn from the Department of Health and Human Services and measured in months, with the full schedule in Table A5 and its geography in Figure A9. The caregiving account predicts that harm should shrink as the exemption lengthens.

[Table 7 About Here]

That is what the estimates show, and they show it for the traits that carry the baseline result. Across all states in Panel A, a longer exemption significantly dampens the agreeableness and neuroticism declines, and the full-index estimate moves in the same direction though it is not precisely estimated. Panel B, which drops states at the federal default and those still phasing in their rules, reproduces the pattern, with the agreeableness moderation slightly larger and again significant. The protection concentrates on agreeableness, the trait most robustly affected in the baseline, so the reform's imprint is measurably lighter in the states that shielded early maternal care the longest.

6.4 When Does Exposure Matter?

My baseline treatment measures the share of the conception-to-age-five window spent under reform, but because reform is permanent, a child exposed early is necessarily exposed later as well, so the estimate pools exposure across the whole of childhood. Resolving *when* in childhood exposure matters is therefore both an interpretive and a substantive question. The developmental logic that motivates this paper makes a clear prediction. If noncognitive skills are laid down through the continuity and quality of care in the earliest years, then the harm should be largest for children exposed from birth or before and should fade as the age at first exposure rises, once the sensitive window has already closed, additional exposure should do little.

I test this by replacing the single dose with a series of indicators for the child's age when reform arrived in their birth state, interacted with target status, and I plot the resulting coefficients in Figure 1. The horizontal axis runs from reform already in place before birth, at the left, through reform arriving in early childhood, to reform arriving only in later childhood, at the right. Because exposure is greatest at the left and declines moving rightward, the developmental prediction is a gradient rising from negative values at early ages toward zero at later ages. The comparison group serves as the internal placebo, and age ten is the omitted reference.

[Fig. 1 About Here]

Figure 1 traces exactly this gradient. The estimated effect on the composite index is most negative for children exposed in utero and through the first years of life, around −0.2 to −0.35 SD, and attenuates steadily toward zero as the age at reform rises, reaching effectively zero for children first exposed in later childhood. The linear fit superimposed on the coefficients summarizes the pattern, a positive slope that carries the effect from its early-life trough up to zero. Two features of the figure support the design as much as the interpretation. The coefficients for reform arriving in later childhood, at the right, are flat and centered on zero, indicating that exposure past the early window leaves no imprint. This also provides another layer suggesting no "pretrend". And the coefficients for reform already in place before birth, at the far left, are similarly stable, which is difficult to reconcile with the effect being generated by cohort trends within states rather than by exposure itself.

6.5 Benchmarking the Effect Size

How large is the estimated decline? A composite effect of 0.23 SD moves a fully exposed target child from roughly the median to the 41st percentile of the NCS distribution, and the trait-specific declines, 0.31 SD for agreeableness and 0.44 SD for openness, are larger still. Because NCS are measured with different instruments, at different ages, and after very different early-life shocks, no comparison is exact, but a handful of estimates place the effect in a reasonable range.

The closest comparisons involve the same traits. Akee et al. (2018), studying an unconditional cash transfer to Native American households, find effects on the three Big Five traits they measure, agreeableness, conscientiousness, and neuroticism, on the order of 0.21 to 0.27 SD, close to my own magnitudes but reversed in direction, since an inflow of early-life resources raised these traits while a reform that tightened assistance lowered them. Openness shows the same pattern: García et al. (2023) find that the Perry Preschool Project raised it by 0.32 SD, a same-trait, opposite-sign counterpart to my 0.44 SD decline.

Two of these estimates also speak to persistence, which matters because I measure skills in young adulthood, once personality has consolidated. The Perry openness effect is observed at age 54, and Attanasio et al. (2024) find that kangaroo mother care reduces externalizing behavior by 0.18 SD fully twenty years after treatment. That effects of this magnitude remain detectable decades later supports reading my estimate as a lasting imprint on the adult trait rather than a transient childhood change.

The direction of my result matches the smaller set of shocks that, like welfare reform, displaced early parental care. Baker et al. (2019) find that expanded subsidized childcare in Quebec worsened children's socioemotional outcomes by 0.13 to 0.27 SD, and Morando and Platt (2022) find that center-based care in infancy raises externalizing behavior by about 0.11 SD. Both fall in a comparable range and point the same way, reinforcing that early shifts away from parental care can depress socioemotional development. Across these cash-transfer, preschool, and childcare shocks, effects on the order of 0.2 to 0.4 SD are typical whether early-life resources are added or withdrawn, and they persist into adulthood. My estimate sits squarely within that range.

6.6 Robustness Checks

In this section, I subject the baseline estimate to the battery of checks described in Section 4. I begin with the two most central to the design, the parallel-trends assumption and the exogeneity of reform timing, and then turn to selective migration, negative weighting, and two checks on unobserved confounding.

The identifying assumption requires that, absent reform, the target and comparison groups would have followed similar trajectories across birth cohorts. I assess this with the event-study specification of Section 4, which replaces the single dose with a series of coefficients by year relative to state reform adoption, omitting the year before adoption as the reference. Figure 2 plots these estimates for the composite index. The pre-adoption coefficients are small and statistically indistinguishable from zero, with a joint test yielding $p = 0.166$, so I find no evidence of differential trends before reform took effect. The effect emerges only from adoption onward; the pattern expected of a treatment that begins with exposure rather than a pre-existing divergence. Because every state in the sample adopts reform within a narrow window and none remains untreated, I confirm this in the stacked event study of Figure A7, in which each adoption cohort is matched to not-yet-treated states and the sub-experiments are stacked with cohort-specific fixed effects. The stacked estimates reproduce the same flat pre-period, with a joint pre-adoption test of $p = 0.275$, so the result does not depend on the comparisons that generate negative weights under staggered adoption.

[Figure 2 About Here]

A second concern is that states timed reform in response to economic or social conditions that also bore on children's development. Table 11 examines this by regressing the timing of state reform adoption, measured as months from January 1992 to the earlier of a state's waiver or TANF date, on a vector of pre-reform state characteristics, across the full universe of reform states and in specifications that add region fixed effects, weight by population, and exclude influential states. In the baseline column, no individual characteristic is a significant predictor of adoption timing, and the pre-reform characteristics jointly explain little of the variation in timing, with an R-squared of 0.109. This weak fit is itself reassuring for the design, echoing the finding of Ziliak et al. (2000) that reform timing was largely unrelated to states' pre-reform economic conditions. The pattern holds across the remaining specifications, where no characteristic emerges as a consistent predictor. The absence of systematic determinants supports the assumption that reform timing is plausibly exogenous to the state conditions relevant for child development, and these characteristics enter the main specification interacted with birth cohort, absorbing any residual correlation directly.

[Table 11 About Here]

Because exposure is assigned by state of birth, families who moved during the exposure window could break the link between assigned and realized policy exposure. Table 12 addresses this by restricting the sample to children who remained in their birth state through early childhood, for whom the assigned exposure reflects the policy environment they actually experienced, and building the specification up sequentially from fixed effects through the full control set. The interaction is stable throughout, at -0.29 in the preferred column against −0.23 in the full sample, so restricting to children whose early-life exposure is measured without migration error, if anything, strengthens the estimate, and selective migration does not account for the effect.

[Table 12 About Here]

Next, since states adopt reform at different times and the effect may vary across cohorts and states, the standard two-way fixed-effects estimator can place negative weights on some comparisons, allowing already-treated cohorts to serve as controls for later-treated ones. I gauge the severity of this in Table 8, decomposing the estimand into its 104-underlying group-by-period effects following de Chaisemartin and d'Haultfoeuille (2020). Only 33 receive negative weights, and these sum to −0.10, about a tenth of the total weight mass, while the sensitivity measures indicate that the estimate would change sign only under treatment-effect heterogeneity far larger than is plausible here. I then estimate the stacked design of Table 9, in which each adoption cohort is matched to not-yet-treated states and the sub-experiments are stacked with cohort-specific fixed effects, so that identification never draws on forbidden comparisons. The stacked composite estimate is −0.21 and significant, close to the baseline, and the subscale pattern is preserved, with agreeableness at −0.26 and openness at −0.62, both significant. That the effect survives a design purged of negative-weight comparisons indicates it is not an artifact of the staggered rollout.

[Table 9 About Here]

Also, because the exogeneity of timing cannot be established conclusively through covariate balance alone, I complement the timing regression with a non-parametric randomization test that probes directly whether the baseline estimate could arise by chance, in the spirit of Ferrara et al. (2012). The logic is straightforward. If treatment eligibility carries a real signal, then randomly reassigning target status across individuals should sever any systematic relationship between early-life exposure and young-adult NCS. For each permutation r, I draw a placebo target indicator $T_i^{(r)}$, form the corresponding placebo interaction $RE_{isc}^{\text{IU-5}} \times T_i^{(r)}$, re-estimate the baseline specification, and store the coefficient $\widehat{\beta}^{(r)}$. Repeating this 1,000 times yields an empirical null distribution $\left\{\widehat{\beta}^{(r)}\right\}_{r=1}^{R}$ under the hypothesis of no effect. The randomization p-value is the share of placebo estimates at least as large in magnitude as the observed coefficient,

$$p = \frac{1}{R} \sum_{r=1}^{R} \mathbf{1} \left(\left| \widehat{\beta}^{(r)} \right| \geq \left| \widehat{\beta} \right| \right).$$

Figure 3 plots the resulting distribution against the observed estimate. The placebo coefficients center on zero, as expected when eligibility is assigned at random, and the observed effect of −0.23 lies in the extreme left tail, yielding an empirical p-value of 0.002. Because the test is distribution-free, it does not rely on the asymptotic approximation behind the clustered standard errors, and together with the timing regression it provides a strong final check on identification, the one evaluating whether reform timing is predictable, the other whether the observed effect exceeds what arbitrary assignment of eligibility would produce.

[Figure 3 About Here]

Finally, I bound the scope for omitted-variable bias following Oster (2019). Table 13 reports, for each outcome, the coefficient's movement as controls are added and the degree of selection on unobservables, relative to selection on the included controls, that would be required to drive the estimate to zero. For the composite index the coefficient is essentially unchanged as controls enter, moving from −0.217 to −0.226, and the implied δ is 5.83, meaning unobservables would need to be more than five times as important as the full set of observed controls to explain away the effect. Agreeableness is similarly stable with a δ of 1.72. The identified set excludes zero for the composite and for agreeableness, so both the coefficient stability and the Oster bounds point to an effect that is robust to plausible unobserved confounding.

[Table 13 About Here]

6.7 Labor-Market Returns to NCS

The estimates so far show that welfare reform lowered NCS, with the decline concentrated in agreeableness and openness. I next ask whether these socioemotional changes are reflected in adult economic outcomes. I do so in two steps. First, I estimate the association between NCS and adult labor-market outcomes in an augmented Mincer framework.

Second, I estimate the reduced-form effect of early-life reform exposure on adult earnings and employment.

Table 10 reports the returns to skill. Columns 1 and 4 present the baseline human-capital specification, while columns 2 and 5 add the composite NCS index. The composite index is not significantly related to earnings, with a coefficient of -0.04. The disaggregated estimates, however, reveal substantial differences across traits. Agreeableness is associated with lower earnings, with a coefficient of -0.21, and openness is also negatively associated with earnings, at -0.11. By contrast, conscientiousness is positively associated with earnings, while neuroticism has a smaller positive association. The composite null therefore masks offsetting associations across traits. A similar pattern emerges for employment, which is more strongly related to conscientiousness and neuroticism and is not meaningfully related to agreeableness or openness.

This pricing pattern helps interpret the reduced-form labor-market estimates. Welfare reform reduced agreeableness and openness, the two traits that are not positively rewarded in the earnings regressions. A mechanical decomposition combining the estimated skill changes with these descriptive returns would therefore not predict an earnings loss and may even imply a gain. The reduced-form estimates, however, show neither a gain nor a loss. Table A3 reports no detectable effect of early-life reform exposure on adult earnings or employment. The null earnings effect is consistent with prior evidence finding no detectable effect of welfare reform exposure on adult earnings (Hartley et al., 2022).

The absence of detectable labor-market effects does not imply that the NCS losses are inconsequential. Rather, the traits affected by the reform are not those most strongly rewarded in earnings or employment. The developmental consequences of welfare reform therefore appear on socioemotional margins that conventional labor-market outcomes may not fully capture.

7 Discussion and Conclusion

This study asks whether early-life exposure to welfare reform had lasting effects on the socioemotional development of the next generation. Using variation in reform timing across states and birth cohorts, together with differences in families' likelihood of being affected, I find that exposure from in utero through age five reduced NCS in young adulthood. The effects are concentrated among children born to mothers with a high school education or less, the group most likely to have been directly affected by the reform. The decline is driven primarily by agreeableness and openness, although the estimate for the latter is less precise. These findings show that welfare reform affected not only contemporaneous family outcomes, but also a dimension of human capital that remained visible as exposed children entered adulthood.

My findings add a longer-run perspective to a literature that has produced mixed evidence on children's behavioral and socioemotional responses to welfare reform. Earlier welfare experiments found improvements in some behavioral outcomes when work requirements were paired with earnings supplements, while other studies linked reform to increased adolescent delinquency and risky behavior (Gennetian et al., 2004; Chase-Lansdale et al., 2011; Dave et al., 2021). Other work finds little evidence of lasting socioemotional effects (Herbst, 2017; Vaughn, 2022). I do not view these findings as necessarily contradictory. Much of the earlier evidence captures behavior while children and adolescents are still responding to contemporaneous changes in maternal work, supervision, and household conditions. I instead measure socioemotional traits in young adulthood, when these traits are more stable and may better reveal whether early exposure left a persistent developmental imprint (Bleidorn et al., 2022).

The age-at-exposure pattern provides the clearest clue to how that imprint may have emerged. I find the largest adverse effects among children exposed in utero and during the first years of life, with the estimates progressively attenuating as age at first exposure increases. This gradient is consistent with evidence that conditions experienced during sensitive periods early in life can have consequences that persist well beyond the initial exposure (Currie and Almond, 2011). It also fits models of skill formation in which early capabilities shape later development through self-productivity and dynamic complementarity (Cunha and Heckman, 2007; Heckman and Mosso, 2014). I therefore interpret the age gradient not simply as heterogeneity in the treatment effect, but as evidence that the timing of the developmental environment altered by welfare reform matters for understanding the long-run decline in NCS.

The mechanism estimates provide suggestive evidence on what may have changed during this sensitive period. Because these measures capture caregiving conditions at a single point in time, I interpret them cautiously. I find little evidence of gains in household income or maternal employment, but suggestive changes in formal childcare use and household composition, together with a negative, though imprecisely estimated, effect on mother-child closeness. These patterns point to changes in the organization and relational quality of early care. The presence of younger children can increase caregiving demands and redistribute parental time and attention within the household (Gennetian et al., 2004; Price, 2008), while greater reliance on formal childcare changes both the setting and nature of children's interactions with caregivers. These margins may be especially consequential during the first years of life, when sustained one-to-one adult interaction is an important developmental input (Fort et al., 2020). Evidence on center-based care and large childcare expansions further suggests that the type and quality of substitute care can shape behavioral and persistent NCS outcomes (Morando and Platt, 2022; Baker et al., 2019). I therefore view the mechanism evidence as suggestive, but broadly consistent with welfare reform altering early caregiving environments.

The concentration of the decline in agreeableness is particularly consistent with this caregiving pathway. Agreeableness captures prosocial motivation, interpersonal responsiveness, and emotion regulation in social interactions, with its developmental origins traced to early cooperation, self-control, and parenting experiences (Baardstu et al., 2017). The decline in openness is also directionally consistent with disruptions to early environments that support curiosity, exploration, and engagement with new experiences, although this estimate is less precise. I therefore view the trait-specific results as broadly consistent with welfare reform altering developmental inputs during a period when these socioemotional capacities remain malleable.

The AYCE estimates provide a further test of this interpretation. If changes in caregiving during the earliest years of life contributed to the decline in NCS, I should find weaker effects where mothers of very young children were protected from immediate work requirements. That is what I find. Longer exemptions attenuate the adverse effects on NCS, with the clearest moderation for agreeableness. This pattern complements Herbst (2017), who shows that variation in these exemptions altered children's early developmental outcomes. I do not interpret this evidence as showing that maternal employment or formal childcare is inherently harmful. Rather, the AYCE results reinforce the broader pattern in my estimates: the developmental consequences of welfare reform depended importantly on whether its work requirements reached families during the earliest years of a child's life.

The heterogeneity results are informative but warrant caution. I estimate larger and more precise effects among girls, although the interaction tests do not provide strong evidence that girls were affected differently from boys. I therefore interpret this pattern as showing where the effects are most detectable rather than as evidence of a distinct gender response to welfare reform. This interpretation is consistent with evidence that boys and girls can respond differently to early family environments and disadvantage (Autor et al., 2019). A similar caution applies to race. The effects are more visible among White children, but the differences across racial groups are imprecisely estimated. Importantly, splitting the sample substantially reduces precision and limits the power of these comparisons. I therefore do not interpret the estimates as evidence of structural racial differences in treatment effects. They may instead reflect differences in exposure, policy implementation, baseline environments, or simply statistical precision.

The labor-market results help clarify the nature of the developmental effects I document. The traits most affected by welfare reform, agreeableness and openness, are not the traits most strongly rewarded in earnings or employment. In my descriptive returns estimates, conscientiousness and emotional stability are more strongly associated with favorable labor-market outcomes, while agreeableness and openness are weakly or negatively associated with these outcomes. This pattern is consistent with evidence that labor-market returns vary considerably across personality traits, including limited or negative earnings returns to agreeableness (Flinn et al., 2025). It is therefore not surprising that I find no detectable reduced-form effects on adult earnings or employment. I do not interpret these null effects as evidence that the NCS losses are inconsequential. Rather, they suggest that the affected traits may matter along dimensions of adult well-being that earnings and employment do not fully capture.

The broader implication is that the consequences of welfare reform extend beyond its immediate effects on adult employment, income, and program participation. Policies that reshape the constraints facing parents can also alter the developmental environments of their children, with effects that remain visible decades later. My findings therefore underscore the importance of developmental timing and intergenerational outcomes in evaluating safety-net policy, particularly when conventional economic outcomes may not capture its full long-run consequences.

References

Akee, R., Copeland, W., Costello, E. J., and Simeonova, E. (2018). How does household income affect child personality traits and behaviors? *American Economic Review*, 108(3):775–827.

Attanasio, O., Cortes, D., Maldonado, D., Rodriguez-Lesmes, P., Charpak, N., Tessier, R., Ruiz, J. G., Gallego, J., Hernandez, T., Uriza, F., et al. (2024). Parental investments and skills formation during infancy and youth: Long term evidence from an early childhood intervention. Technical report, National Bureau of Economic Research.

Aughinbaugh, A. and Gittleman, M. (2004). Maternal employment and adolescent risky behavior. *Journal of Health Economics*, 23(4):815–838.

Autor, D., Figlio, D., Karbownik, K., Roth, J., and Wasserman, M. (2019). Family disadvantage and the gender gap in behavioral and educational outcomes. *American Economic Journal: Applied Economics*, 11(3):338–381.

Baardstu, S., Karevold, E. B., and von Soest, T. (2017). Childhood antecedents of agreeableness: A longitudinal study from preschool to late adolescence. *Journal of Research in Personality*, 67:202–214.

Baker, M., Gruber, J., and Milligan, K. (2019). The long-run impacts of a universal child care program. *American Economic Journal: Economic Policy*, 11(3):1–26.

Bastian, J., Bian, L., and Grogger, J. (2021). How did safety-net reform affect early adulthood among adolescents from low-income families? *National Tax Journal*, 74(3):825–865.

Benjamini, Y., Krieger, A. M., and Yekutieli, D. (2006). Adaptive linear step-up procedures that control the false discovery rate. *Biometrika*, 93(3):491–507.

Bennhoff, F. H., García, J. L., and Leaf, D. E. (2024). The dynastic benefits of early-childhood education: Participant benefits and family spillovers. *Journal of Human Capital*, 18(1):44–73.

Bitler, M. and Hoynes, H. W. (2010). The state of the safety net in the post-welfare reform era. NBER Working Paper w16504, National Bureau of Economic Research.

Blank, R. M. (2002). Evaluating welfare reform in the united states. *Journal of Economic Literature*, 40(4):1105–1166.

Blank, R. M. (2009). What we know, what we don't know, and what we need to know about welfare reform. In *Welfare Reform and Its Long-Term Consequences for America's Poor*, pages 22–58. Cambridge University Press.

Bleidorn, W., Schwaba, T., Zheng, A., Hopwood, C. J., Sosa, S. S., Roberts, B. W., and Briley, D. A. (2022). Personality stability and change: A meta-analysis of longitudinal studies. *Psychological Bulletin*, 148(7-8):588.

Bolbocean, C. and Tylavsky, F. A. (2021). The impact of safety net programs on early-life developmental outcomes. *Food Policy*, 100:102018.

Callaway, B., Goodman-Bacon, A., and Sant'Anna, P. H. (2024). Difference-in-differences with a continuous treatment. Technical report, National Bureau of Economic Research.

Cengiz, D., Dube, A., Lindner, A., and Zipperer, B. (2019). The effect of minimum wages on low-wage jobs. *The quarterly journal of economics*, 134(3):1405–1454.

Chase-Lansdale, P. L., Cherlin, A. J., Guttmannova, K., Fomby, P., Ribar, D. C., and Coley, R. L. (2011). Long-term implications of welfare reform for the development of adolescents and young adults. *Children and Youth Services Review*, 33(5):678–688.

Chyi, H. and Ozturk, O. D. (2013). The effects of single mothers' welfare use and employment decisions on children's cognitive development. *Economic Inquiry*, 51(1):675–706.

Chyi, H., Ozturk, O. D., and Zhang, W. (2014). Welfare reform and children's early cognitive development. *Contemporary Economic Policy*, 32(4):729–751.

Corman, H., Dave, D., Kalil, A., and Reichman, N. E. (2017a). Effects of maternal work incentives on youth crime. *Labour Economics*, 49:128–144.

Corman, H., Dave, D. M., Kalil, A., and Reichman, N. E. (2017b). Effects of maternal work incentives on teen drug arrests. Working paper.

Corman, H., Dave, D. M., Schwartz-Soicher, O., and Reichman, N. E. (2022). Effects of welfare reform on household food insecurity across generations. *Economics & Human Biology*, 45:101101.

Crouse, G. (1999). State implementation of major changes to welfare policies, 1992–1998. Technical report, U.S. Department of Health and Human Services, Washington, DC.

Cunha, F. and Heckman, J. (2007). The technology of skill formation. *American Economic Review*, 97(2):31–47.

Currie, J. and Almond, D. (2011). Human capital development before age five. In *Handbook of Labor Economics*, volume 4, pages 1315–1486. Elsevier.

Dave, D., Corman, H., Kalil, A., Schwartz-Soicher, O., and Reichman, N. E. (2021). Intergenerational effects of welfare reform: Adolescent delinquent and risky behaviors. *Economic Inquiry*, 59(1):199–216.

Dave, D. M., Corman, H., and Reichman, N. E. (2012). Effects of welfare reform on education acquisition of adult women. *Journal of Labor Research*, 33(2):251–282.

de Chaisemartin, C. and d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.

Deming, D. J. (2017). The growing importance of social skills in the labor market. *Quarterly Journal of Economics*, 132(4):1593–1640.

Dore, E. C., Hamad, R., Komro, K. A., and Livingston, III, M. D. (2025). The long-term health effects of welfare reform. *Social Science & Medicine*, 371:117878.

Duncan, G. J., Huston, A. C., and Weisner, T. S. (2007). *Higher Ground: New Hope for the Working Poor and Their Children*. Russell Sage Foundation.

Dunifon, R., Kalil, A., and Danziger, S. K. (2003). Maternal work behavior under welfare reform: How does the transition from welfare to work affect child development? *Children and Youth Services Review*, 25(1-2):55–82.

Esping-Andersen, G. and Cimentada, J. (2018). Ability and mobility: The relative influence of skills and social origin on social mobility. *Social Science Research*, 75:13–31.

Ferrara, E. L., Chong, A., and Duryea, S. (2012). Soap operas and fertility: Evidence from brazil. *American Economic Journal: Applied Economics*, 4(4):1–31.

Fletcher, J. M. and Wolfe, B. (2016). The importance of family income in the formation and evolution of non-cognitive skills in childhood. *Economics of Education Review*, 54:143–154.

Flinn, C. J., Todd, P. E., and Zhang, W. (2025). Labor market returns to personality: A job search approach to understanding gender gaps. *Journal of Political Economy*, 133(4):1169–1234.

Fort, M., Ichino, A., and Zanella, G. (2020). Cognitive and noncognitive costs of day care at age 0–2 for children in advantaged families. *Journal of Political Economy*, 128(1):158–205.

García, J. L., Heckman, J. J., and Ronda, V. (2023). The lasting effects of early-childhood education on promoting the skills and social mobility of disadvantaged african americans and their children. *Journal of Political Economy*, 131(6):1477–1506.

Gennetian, L. A., Duncan, G., Knox, V., Vargas, W., Clark-Kauffman, E., and London, A. S. (2004). How welfare policies affect adolescents' school outcomes: A synthesis of evidence from experimental studies. *Journal of Research on Adolescence*, 14(4):399–423.

Gennetian, L. A. and Miller, C. (2002). Children and welfare reform: A view from an experimental welfare program in minnesota. *Child Development*, 73(2):601–620.

Goldberg, L. R. (1993). The structure of phenotypic personality traits. *American Psychologist*, 48(1):26.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.

Grogger, J. (2003). The effects of time limits, the eitc, and other policy changes on welfare use, work, and income among female-headed families. *Review of Economics and Statistics*, 85(2):394–408.

Grogger, J. T. and Karoly, L. A. (2005). *Welfare Reform: Effects of a Decade of Change*. Harvard University Press.

Hartley, R. P., Lamarche, C., and Ziliak, J. P. (2022). Welfare reform and the intergenerational transmission of dependence. *Journal of Political Economy*, 130(3):523–565.

Heckman, J. J. (2000). Policies to foster human capital. *Research in Economics*, 54(1):3–56.

Heckman, J. J. (2008). The case for investing in disadvantaged young children. *CESifo DICE Report*, 6(2):3–8.

Heckman, J. J. and Mosso, S. (2014). The economics of human development and social mobility. *Annual Review of Economics*, 6(1):689–733.

Heflin, C. M. and Acevedo, S. K. (2011). Non-income effects of welfare receipt on early childhood cognitive scores. *Children and Youth Services Review*, 33(5):634–643.

Herbst, C. M. (2017). Are parental welfare work requirements good for disadvantaged children? evidence from age-of-youngest-child exemptions. *Journal of Policy Analysis and Management*, 36(2):327–357.

Hofferth, S. L. (1999). Childcare, maternal employment, and public policy. *The Annals of the American Academy of Political and Social Science*, 563(1):20–38.

Hoynes, H., Schanzenbach, D. W., and Almond, D. (2016). Long-run impacts of childhood access to the safety net. *American Economic Review*, 106(4):903–934.

Humphries, J. E. and Kosse, F. (2017). On the interpretation of non-cognitive skills: What is being measured and why it matters. *Journal of Economic Behavior & Organization*, 136:174–185.

Kaestner, R. and Chan Lee, W. (2005). The effect of welfare reform on prenatal care and birth weight. *Health Economics*, 14(5):497–511.

Kalil, A., Corman, H., Dave, D., Schwarz-Soicher, O., and Reichman, N. E. (2023). Welfare reform and the quality of young children's home environments. *Demography*, 60(6):1791–1813.

Kautz, T., Heckman, J. J., Diris, R., Ter Weel, B., and Borghans, L. (2014). Fostering and measuring skills: Improving cognitive and non-cognitive skills to promote lifetime success. Nber working paper, National Bureau of Economic Research.

Kling, J. R., Liebman, J. B., and Katz, L. F. (2007). Experimental analysis of neighborhood effects. *Econometrica*, 75(1):83–119.

Miller, A. R. and Zhang, L. (2009). The effects of welfare reform on the academic performance of children in low-income households. *Journal of policy Analysis and Management*, 28(4):577–599.

Milligan, K. and Stabile, M. (2011). Do child tax benefits affect the well-being of children? evidence from canadian child benefit expansions. *American Economic Journal: Economic Policy*, 3(3):175–205.

Morando, G. and Platt, L. (2022). The impact of centre-based childcare on non-cognitive skills of young children. *Economica*, 89(356):908–946.

Morris, P., Gennetian, L. A., Duncan, G. J., and Huston, A. (2009). *How Welfare Policies Affect Child and Adolescent School Performance: Investigating Pathways of Influence with Experimental Data*. Cambridge University Press, Cambridge, UK.

Offner, P. (2005). Welfare reform and teenage girls. *Social Science Quarterly*, 86(2):306–322.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204.

Phillips, D. A. and Shonkoff, J. P., editors (2000). *From Neurons to Neighborhoods: The Science of Early Childhood Development*. National Academy Press, Washington, DC.

Pilkauskas, N. V., Brooks-Gunn, J., and Waldfogel, J. (2018). Maternal employment stability in early childhood: Links with child behavior and cognitive skills. *Developmental Psychology*, 54(3):410.

Price, J. (2008). Parent-child quality time: Does birth order matter? *Journal of Human Resources*, 43(1):240–265.

Reichman, N. E., Corman, H., Dave, D., Kalil, A., and Schwartz-Soicher, O. (2023). Effects of welfare reform on positive health and social behaviors of adolescents. *Children*, 10(2):260.

Roodman, D., Nielsen, M. Ø., MacKinnon, J. G., and Webb, M. D. (2019). Fast and wild: Bootstrap inference in stata using boottest. *The Stata Journal*, 19(1):4–60.

Schoeni, R. F. and Blank, R. M. (2000). What has welfare reform accomplished? impacts on welfare participation, employment, income, poverty, and family structure. Working paper, National Bureau of Economic Research.

UKCPR (2026). UKCPR National Welfare Data, 1980–2023. University of Kentucky Center for Poverty Research, Gatton College of Business and Economics, University of Kentucky, Lexington, KY.

Vaughn, C. (2022). Long-run impact of welfare reform on educational attainment and family structure. Available at SSRN 4131589.

Vaughn, C. N. (2023). Welfare reform and childhood health status and utilization. *Children and Youth Services Review*, 150:107004.

Walker, S. P., Chang, S. M., Wright, A. S., Pinto, R., Heckman, J. J., and Grantham-McGregor, S. M. (2022). Cognitive, psychosocial, and behaviour gains at age 31 years from the jamaica early childhood stimulation trial. *Journal of Child Psychology and Psychiatry*, 63(6):626–635.

Wing, C., Freedman, S. M., and Hollingsworth, A. (2024). Stacked difference-in-differences. Technical report, National Bureau of Economic Research.

Wolfers, J. (2006). Did unilateral divorce laws raise divorce rates? a reconciliation and new results. *American Economic Review*, 96(5):1802–1820.

Ziliak, J. P. (2015). Temporary assistance for needy families. In *Economics of Means-Tested Transfer Programs in the United States, Volume 1*, pages 303–393. University of Chicago Press.

Ziliak, J. P., Figlio, D. N., Davis, E. E., and Connolly, L. S. (2000). Accounting for the decline in afdc caseloads: Welfare reform or the economy? *Journal of Human Resources*, pages 570–586.

Tables and Figures

Table 1: Summary Statistics

	N	Mean	SD
Outcomes			
Noncognitive Skills Index	2837	0.009	0.420
Agreeableness	2837	-0.004	0.713
Conscientiousness	2837	-0.009	0.732
Extraversion	2837	0.016	0.753
Neuroticism	2837	0.042	0.794
Openness	2837	0.001	0.789
Treatment			
Exposure (IU-Age 5)	2837	0.306	0.316
Target	2837	0.153	0.360
Young Adult Controls			
Age at Survey	2837	23.435	2.723
Female	2837	0.487	0.500
Nonwhite	2837	0.266	0.442
Grandparent Present	2837	0.024	0.154
Maternal Controls			
Mother's Age	2837	34.145	7.443
< High School	2837	0.151	0.358
High School	2837	0.312	0.464
Some College	2837	0.460	0.499
Pre-Reform State Level Characteristics			
Unemployment Rate	2837	6.314	1.368
ln(TANF Benefit)	2837	12.483	0.970
ln(SNAP Benefit)	2837	13.104	0.836
Poverty Rate	2837	13.511	3.636
Minimum Wage	2837	3.757	0.560
EITC Rate	2837	0.011	0.067

Notes: Survey-weighted summary statistics for the analytic sample, drawn from the PSID Child Development Supplement and Transition into Adulthood. NCS measures are standardized z-scores.

Table 2: Baseline Estimation

	(1)	(2)	(3)	(4)	(5)
IU-Age 5	-0.031	-0.033	-0.031	-0.048	-0.053
	(0.103)	(0.103)	(0.104)	(0.099)	(0.104)
IU-Age 5 × Target	-0.234***	-0.228***	-0.226***	-0.226***	-0.227***
	(0.083)	(0.080)	(0.078)	(0.081)	(0.082)
Bootstrap p-value	0.012	0.015	0.011	0.014	0.018
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Mother Controls	No	Yes	Yes	Yes	Yes
Young Adult Controls	No	No	Yes	Yes	Yes
State Controls	No	No	No	Yes	Yes
State Linear Trends	No	No	No	No	Yes
Observations	2837	2837	2837	2837	2837
Adj. R-squared	0.047	0.050	0.075	0.085	0.097

Notes: This table reports difference-in-difference-in-differences (DDD) estimates of the effect of early-life welfare reform exposure on the standardized noncognitive skills (NCS) index measured in young adulthood. "IU-Age 5" is the fraction of the conception-to-age-5 window occurring after welfare reform, and "Target" identifies children born to mothers with a high school education or less. IU-Age 5 × Target is the coefficient of interest. Column (1) includes birth-cohort, survey-year, target, and state fixed effects. Columns (2)–(4) progressively add maternal characteristics, young-adult characteristics, and pre-reform state economic and policy controls. Column (5) adds state-specific linear birth-cohort trends; column (4) is the preferred specification. Estimates are survey weighted. Standard errors clustered at the state level are reported in parentheses. Bootstrap p-values for the DDD coefficient use wild cluster bootstrap inference with 9,999 replications and Rademacher weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Big Five Subscales

	(1) Agreeableness	(2) Conscientiousness	(3) Extraversion	(4) Neuroticism	(5) Openness
IU-Age 5	0.007	-0.182	-0.229	-0.164	0.332*
	(0.140)	(0.146)	(0.138)	(0.188)	(0.186)
IU-Age 5 × Target	-0.314**	-0.162	-0.073	-0.140	-0.437**
	(0.130)	(0.153)	(0.172)	(0.147)	(0.192)
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Mother Controls	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes
Unadjusted p	0.021	0.296	0.674	0.346	0.029
BKY q-value	0.044	0.296	0.674	0.346	0.044
Bootstrap *p*-value	0.030	0.317	0.679	0.397	0.034
Observations	2837	2837	2837	2837	2837

Notes: This table reports difference-in-difference-in-differences (DDD) estimates for each of the five Big Five personality subscales, measured as standardized z-scores in young adulthood. "IU-Age 5" measures the fraction of the conception-to-age-5 window occurring under post-reform rules; "Target" identifies children born to mothers with at most a high school education or less. The interaction IU-Age 5 × Target is the causal DDD estimate. All columns include maternal, young adult, and pre-reform state economic controls, together with birth-cohort, survey-year, target, and state fixed effects. To account for testing five outcomes simultaneously, I report multiple-hypothesis-adjusted p-values for the interaction term: BKY denotes the Benjamini-Krieger-Yekutieli (2006) adaptive two-stage q-values, controlling the false discovery rate at q = 0.10. Bootstrap *p*-values denote wild cluster bootstrap values for the interaction term (9,999 replications, Rademacher weights). Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Gender Heterogeneity — Full Index and Subscales

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Index	Agreeableness	Conscientiousness	Extraversion	Neuroticism	Openness
IU-Age 5 (Male)	-0.041	-0.016	-0.185	-0.085	-0.041	-0.016
	(0.100)	(0.161)	(0.129)	(0.194)	(0.100)	(0.161)
IU-Age 5 × Target (Male)	-0.170	-0.249	0.073	-0.005	-0.170	-0.249
	(0.137)	(0.191)	(0.226)	(0.242)	(0.137)	(0.191)
Female	-0.024	0.131**	0.059	0.256***	-0.024	0.131**
	(0.045)	(0.060)	(0.063)	(0.080)	(0.045)	(0.060)
IU-Age 5 × Female	-0.015	0.046	0.004	-0.295	-0.015	0.046
	(0.097)	(0.108)	(0.138)	(0.215)	(0.097)	(0.108)
IU-Age 5 × Target × Female	-0.092	-0.119	-0.398	-0.068	-0.092	-0.119
	(0.162)	(0.224)	(0.288)	(0.223)	(0.162)	(0.224)
Female Total Effect	-0.262	-0.367	-0.326	-0.074	-0.262	-0.367
SE	0.095	0.156	0.208	0.168	0.095	0.156
p-value	0.009	0.024	0.126	0.663	0.009	0.024
Mother Controls	Yes	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes	Yes
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2837	2837	2837	2837	2837	2837
Adj. R-squared	0.084	0.073	0.045	0.057	0.084	0.073

Notes: This table reports difference-in-difference-in-differences (DDD) estimates allowing the effect of welfare reform to vary by child gender, for the composite NCS index and each Big Five subscale (all standardized). The reference group is male children: "IU-Age 5 × Target (Male)" is the DDD estimate for boys, and the triple interaction "IU-Age 5 × Target × Female" is the additional effect for girls relative to boys. "Female Total Effect" reports the implied DDD estimate for girls (the sum of the male effect and the female differential), with its standard error and p-value obtained by linear combination. All specifications include maternal characteristics (education, age), young adult controls (age, race, grandparent caregiving), and pre-reform state economic conditions (unemployment rate, log TANF and SNAP benefits, poverty rate, minimum wage, EITC rate), together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 5: Race Heterogeneity — Full Index and Subscales

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Index	Agreeableness	Conscientiousness	Extraversion	Neuroticism	Openness
IU-Age 5 (White)	-0.011	0.046	-0.172	-0.208	-0.136	0.422*
	(0.091)	(0.123)	(0.139)	(0.130)	(0.178)	(0.212)
IU-Age 5 × Target (White)	-0.293**	-0.467*	-0.010	-0.262	-0.115	-0.608**
	(0.128)	(0.255)	(0.201)	(0.256)	(0.223)	(0.288)
Nonwhite	0.175***	0.063	0.063	0.070	0.344***	0.331***
	(0.038)	(0.057)	(0.063)	(0.092)	(0.051)	(0.078)
IU-Age 5 × Nonwhite	-0.173*	-0.234*	-0.029	-0.054	-0.198	-0.345*
	(0.095)	(0.124)	(0.130)	(0.202)	(0.183)	(0.195)
IU-Age 5 × Target × Nonwhite	0.216	0.418	-0.207	0.335	0.091	0.443
	(0.148)	(0.289)	(0.241)	(0.257)	(0.235)	(0.369)
Nonwhite Total Effect	-0.076	-0.049	-0.217	0.073	-0.023	-0.165
SE	0.102	0.147	0.185	0.231	0.182	0.256
p-value	0.458	0.738	0.249	0.753	0.899	0.524
Mother Controls	Yes	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes	Yes
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2837	2837	2837	2837	2837	2837
Adj. R-squared	0.087	0.078	0.043	0.054	0.218	0.075

Notes: This table reports difference-in-difference-in-differences (DDD) estimates allowing the effect of welfare reform to vary by child race, for the composite NCS index and each Big Five subscale (all standardized). The reference group is White children: "IU-Age 5 × Target (White)" is the DDD estimate for White children, and the triple interaction "IU-Age 5 × Target × Nonwhite" is the additional effect for Nonwhite children relative to White children. "Nonwhite Total Effect" reports the implied DDD estimate for Nonwhite children (the sum of the White effect and the nonwhite differential), with its standard error and p-value obtained by linear combination. All specifications include maternal characteristics (education, age), young adult controls (age, gender, grandparent caregiving), and pre-reform state economic conditions (unemployment rate, log TANF and SNAP benefits, poverty rate, minimum wage, EITC rate), together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Mechanism Analysis

	(1) Informal Childcare	(2) Formal Childcare	(3) Income	(4) Employment	(5) Distress	(6) Young Child	(7) Mother–Child
IU-Age 5	0.083	0.030	3.651	0.121*	-0.313	0.231	-0.038
	(0.052)	(0.045)	(12.588)	(0.063)	(0.817)	(0.440)	(0.070)
IU-Age 5 × Target	-0.055	0.075*	13.592	-0.026	0.845	1.142**	-0.032
	(0.081)	(0.040)	(32.158)	(0.097)	(1.306)	(0.445)	(0.088)
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Maternal controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1863	1863	1868	1868	1144	1868	1863
Adj. R-squared	0.101	0.087	0.085	0.090	0.190	0.643	0.066

Notes: This table reports difference-in-difference-in-differences (DDD) estimates of the effect of early-life welfare-reform exposure on candidate mediating channels measured in the 1997 Child Development Supplement: childcare arrangements (informal, formal), household income, maternal employment, maternal psychological distress, presence of a young child, and mother-child closeness. "IU-Age 5" measures the fraction of the conception-to-age-5 window occurring under post-reform rules; "Target" identifies children born to mothers with at most a high school education or less, and IU-Age 5 × Target is the DDD estimate. All models include maternal, young adult, and pre-reform state economic controls, together with birth-cohort, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 7: Age-of-Youngest-Child Exemption (AYCE)

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Index	Agreeableness	Conscientiousness	Extraversion	Neuroticism	Openness
PANEL A: All States						
IU-Age 5 × Target	-0.233**	-0.664***	0.049	-0.082	-0.360***	-0.098
	(0.096)	(0.113)	(0.166)	(0.263)	(0.126)	(0.149)
IU-Age 5 × Target × Exemption	0.001	0.011***	-0.010	-0.001	0.006*	-0.001
	(0.003)	(0.003)	(0.006)	(0.008)	(0.004)	(0.005)
Observations	2837	2837	2837	2837	2837	2837
Adj. R-squared	0.061	0.054	0.038	0.039	0.154	0.059
PANEL B: Phase-in States Excluded						
IU-Age 5 × Target	-0.244**	-0.684***	0.069	0.016	-0.418***	-0.192
	(0.105)	(0.123)	(0.179)	(0.288)	(0.147)	(0.159)
IU-Age 5 × Target × Exemption	0.002	0.012***	-0.011	-0.004	0.008*	0.002
Observations	2200	2200	2200	2200	2200	2200
Adj. R-squared	0.064	0.058	0.038	0.047	0.165	0.057
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Mother Controls	Yes	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines whether the effect of welfare reform varies with the generosity of a state's age-of-youngest-child work exemption, the age (in months) below which mothers of young children were exempt from AFDC-JOBS work requirements during the waiver era, drawn from the US Department of Health and Human Services. "IU-Age 5 × Target" is the DDD estimate, and "IU-Age 5 × Target × Exemption" is its interaction with the exemption threshold; a positive interaction indicates that more protective (higher-threshold) states experienced smaller reform-induced harm. Panel A includes all states; Panel B excludes states coded at the federal default and those phasing in their requirements. All models include maternal, young adult, and pre-reform state economic controls, together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses.
* p < 0.10, ** p < 0.05, *** p < 0.01.

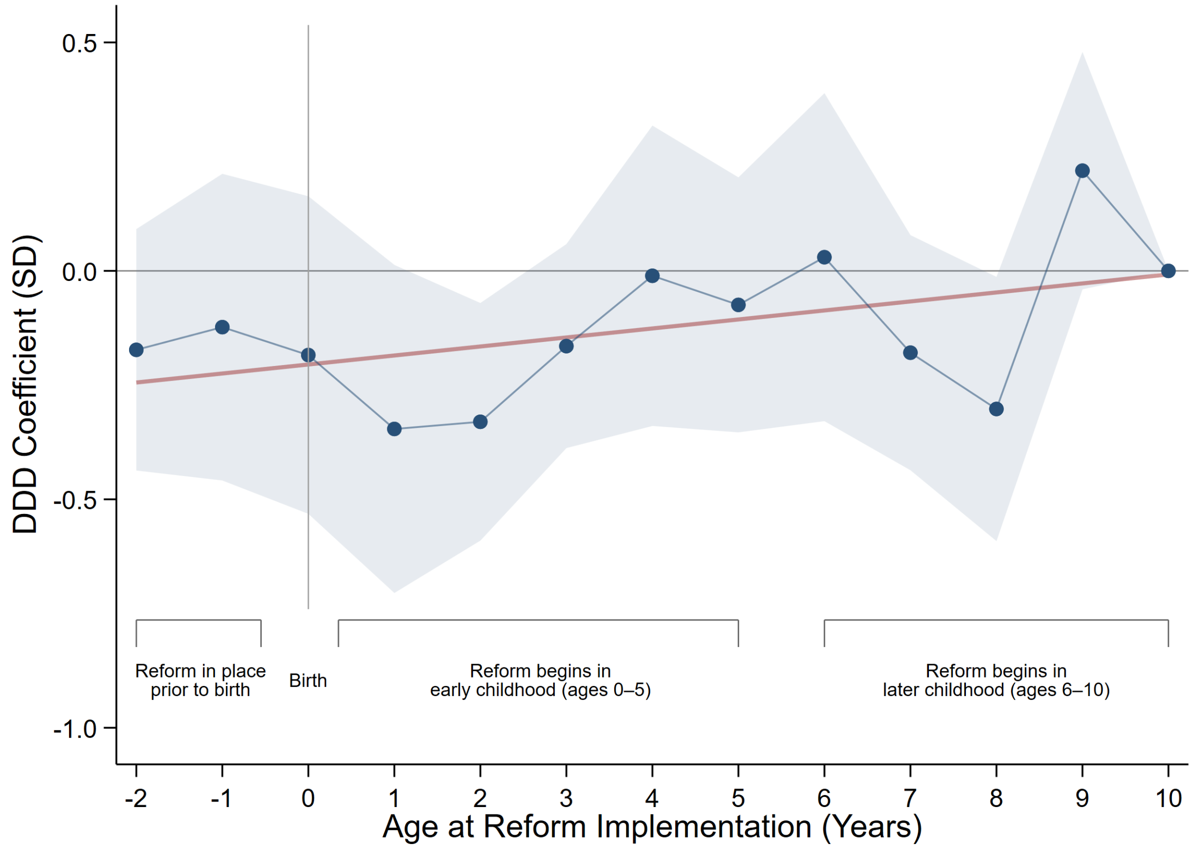

Fig. 1: Effect of Welfare Reform by Age at Exposure
Notes: This figure plots DDD coefficients by the child's age at reform implementation, interacting age-at-reform indicators with target status (age 10 omitted). Because exposure declines from left to right, negative coefficients at younger ages indicate that harm is concentrated among children exposed to reform earliest in life, attenuating toward zero as age at reform rises; the red line is a linear fit summarizing this gradient. The comparison group serves as the internal placebo. The pattern is consistent with a critical-period mechanism in which early-childhood exposure is most consequential. The model includes maternal, young adult, and pre-reform state economic controls, together with survey-year and state fixed effects. Estimates are survey weighted, and shaded bands are 90% confidence intervals from standard errors clustered at the state level.

Table 8: TWFE Weight Decomposition and Sensitivity

Treatment variable: totaltreat05	**ATTs**	**Σ weights**
Positive weights	71	1.1035
Negative weights	33	-0.1035
Total	**104**	**1.0000**

Notes: The two-way fixed-effects estimand is a weighted sum of 104 group-by-period treatment effects; 33 receive negative weights summing to -0.10, roughly a tenth of the total weight mass. The minimum standard deviation of treatment effects required for the estimand and the average treatment effect on the treated to differ in sign is 0.52, and 0.59 for the estimand to have the opposite sign from the effect in every treated cell.

Table 9: Stacked Triple-Difference Estimates: NCS Composite and Big Five Subscales

	(1) Full Index	(2) Agreeableness	(3) Conscientiousness	(4) Extraversion	(5) Neuroticism	(6) Openness
IU-Age 5	-0.073	-0.054	-0.027	-0.250	-0.261*	0.229
	(0.087)	(0.123)	(0.154)	(0.192)	(0.141)	(0.228)
IU-Age 5 × Target	-0.213**	-0.264*	-0.238	0.075	-0.023	-0.620***
	(0.104)	(0.141)	(0.170)	(0.250)	(0.195)	(0.230)
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Survey-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Target FE	Yes	Yes	Yes	Yes	Yes	Yes
Maternal Controls	Yes	Yes	Yes	Yes	Yes	Yes
Young Adult Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10143	10143	10143	10143	10143	10143
Adj. R-squared	0.090	0.078	0.067	0.064	0.224	0.095

Notes: This table reports stacked triple-difference (DDD) estimates of early-life welfare reform exposure on young-adult noncognitive skills, for the composite index and each of the five Big Five subscales. The stacked design forms one sub-experiment per reform-adoption cohort (1992 through 1996). Within each sub-experiment, treated units are children in states adopting reform in the cohort year, and control units are drawn from not-yet-treated states, those adopting strictly later; because every state adopts reform by 1997, no never-treated group exists, so not-yet-treated states serve as the clean controls. This construction confines all comparisons to within a sub-experiment and rules out the forbidden comparisons between earlier- and later-treated units that can generate negative weights under staggered adoption (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfoeuille, 2020). Each sub-experiment is assigned its own state, birth-cohort, survey-year, and target fixed effects, so children near a cohort boundary may enter multiple sub-experiments under distinct references. The coefficient on IU-Age 5 x Target is the DDD effect of a marginal increase in the share of the in-utero to age-five window spent under reform, for children of low-educated single mothers relative to the comparison group. All outcomes are survey weighted using TAS weights, and standard errors, clustered on state, appear in parentheses. NCS measures are standardized z-scores. *** p<0.01, ** p<0.05, * p<0.10.

Table 10: Returns to Skills: Earnings and Employment

	Earnings (PPML)			Employment (LPM)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Full Index	Subscales	Baseline	Full Index	Subscales
Years of Schooling	0.201***	0.201***	0.202***	0.045***	0.044***	0.043***
	(0.031)	(0.031)	(0.030)	(0.009)	(0.009)	(0.009)
Age	1.033***	1.031***	1.073***	0.064	0.073	0.070
	(0.372)	(0.372)	(0.357)	(0.104)	(0.099)	(0.099)
Age squared	-0.021***	-0.021***	-0.021***	-0.001	-0.001	-0.001
	(0.007)	(0.007)	(0.007)	(0.002)	(0.002)	(0.002)
Female = 1	-0.319***	-0.319***	-0.273***	-0.043**	-0.039*	-0.038
	(0.084)	(0.085)	(0.089)	(0.020)	(0.020)	(0.023)
NCS Index (Full)		-0.036			0.118***	
		(0.112)			(0.031)	
Agreeableness			-0.209***			0.024
			(0.059)			(0.015)
Conscientiousness			0.136**			0.037**
			(0.063)			(0.016)
Extraversion			0.025			0.019
			(0.052)			(0.019)
Neuroticism			0.094*			0.026*
			(0.053)			(0.015)
Openness			-0.108**			0.011
			(0.045)			(0.017)
Observations	1853	1853	1853	1854	1854	1854

Notes: This table reports the estimated labor-market returns to skills in adulthood. Columns (1)-(3) model earnings using Poisson pseudo-maximum-likelihood (PPML) on raw earnings; columns (4)-(6) model employment using a linear probability model. Within each outcome, the baseline columns include only schooling and demographics, the second columns add the full standardized NCS index, and the third columns replace it with the five Big Five subscales. All specifications include state, birth-cohort, and survey-year fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Determinants of State Welfare Reform Timing: Robustness Checks

	(1)	(2)	(3)	(4)	(5)
Population	10.534	9.034	19.432	21.156*	36.143
	(11.846)	(12.386)	(12.027)	(11.409)	(23.311)
Unemployment Rate	0.861	0.566	2.505	2.313	4.572
	(2.225)	(2.338)	(2.426)	(2.324)	(4.248)
TANF Receipts	-6.131	-7.398	-10.969	-4.679	-15.429
	(9.456)	(10.094)	(9.664)	(10.455)	(11.171)
SNAP Receipts	-7.170	-4.342	-10.875	-19.285	-21.635
	(14.347)	(15.442)	(15.414)	(15.490)	(21.358)
Poverty Rate	1.092	1.009	1.310	1.439	2.938
	(1.303)	(1.387)	(1.159)	(1.074)	(1.853)
State Minimum Wage	-2.855	-2.818	-6.950	-5.663	-10.080
	(3.947)	(4.161)	(4.734)	(4.629)	(6.175)
State EITC Rate	43.403	43.848	48.933	43.371	96.977
	(33.881)	(33.015)	(37.684)	(38.534)	(95.825)
Region (Ref: Northeast)					
Midwest			-15.242	-9.844	-20.373*
			(9.892)	(9.993)	(11.170)
South			-10.019	-2.092	
			(10.416)	(10.684)	
West			-16.882	-9.120	-24.918*
			(10.262)	(11.270)	(13.662)
Constant	38.671	43.843	18.198	12.666	-66.722
	(64.283)	(66.828)	(59.490)	(56.304)	(123.946)
Observations	50	50	50	48	34
R-squared	0.109	0.108	0.198	0.178	0.279

Notes: This table regresses the timing of state welfare-reform adoption (months from January 1992 to the earlier of a state's waiver or TANF implementation date) on pre-reform state characteristics, to assess whether adoption timing is systematically related to observable economic and demographic conditions. Column (1) is the baseline; column (2) weights by population; column (3) adds Census region fixed effects; column (4) excludes California and New York; column (5) excludes Southern states. Population, TANF receipts, and SNAP receipts enter in logs. The absence of consistently significant predictors supports the identifying assumption that reform timing is plausibly exogenous to state conditions. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

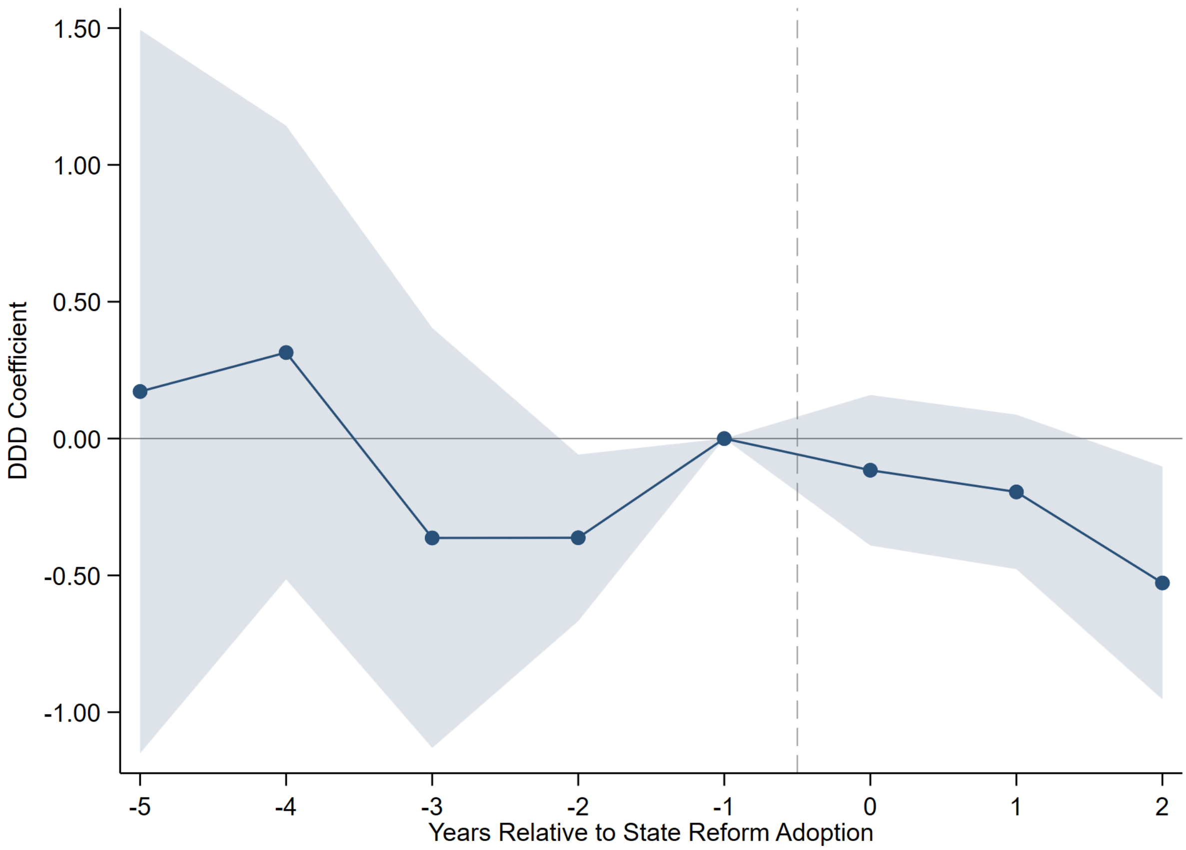

Fig. 2: Event-Study Estimates by Year Relative to State Reform Adoption
Notes: This figure plots the dynamic DDD coefficients from an event-study specification, showing the effect of welfare-reform exposure on the composite NCS index by year relative to state reform adoption. Each point is the interaction of exposure and target with an indicator for a given event year, relative to the omitted reference year ($t = -1$); post-adoption years at $t \geq 2$ are pooled to address thin late-cohort cells. Flat, insignificant pre-period coefficients support the parallel-trends assumption; the joint test of the pre-adoption coefficients yields $p = 0.166$. The model includes maternal, young adult, and pre-reform state economic controls, together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted, and shaded bands are 90% confidence intervals from standard errors clustered at the state level.

Table 12: Migration Robustness — Stayers Only (Composite NCS Index)

	(1)	(2)	(3)	(4)	(5)
IU-Age 5	-0.081	-0.076	-0.056	-0.063	-0.049
	(0.128)	(0.129)	(0.137)	(0.140)	(0.181)
IU-Age 5 × Target	-0.294**	-0.293***	-0.282**	-0.288**	-0.268**
	(0.110)	(0.101)	(0.104)	(0.112)	(0.119)
Birth-Cohort FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Mother Controls	No	Yes	Yes	Yes	Yes
Young Adult Controls	No	No	Yes	Yes	Yes
State Controls	No	No	No	Yes	Yes
State Linear Trends	No	No	No	No	Yes
Observations	1828	1828	1828	1828	1828
Adj. R-squared	0.074	0.082	0.110	0.108	0.122

Notes: This table restricts the sample to stayers, children whose state of birth equals their exposure state, for whom early-life reform exposure is measured without migration error. Columns add controls progressively: column (1) includes only birth-cohort, survey-year, target, and state fixed effects; columns (2)-(4) add maternal characteristics (education, age), young adult controls (age, gender, race, grandparent caregiving), and pre-reform state economic conditions (unemployment rate, log TANF and SNAP benefits, poverty rate, minimum wage, EITC rate); column (5) adds state-specific linear birth-cohort trends. Column (4) is the preferred specification, and the sample is held fixed across columns. The stability of the IU-Age 5 × Target estimate within the stayer sample indicates that the main results are not driven by selective migration into or out of reform states. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Bounds Estimates for Coefficient Stability and Robustness to Omitted Variable Bias

Treatment variable	**(1)** Baseline coeff. β (SE) $[R^2_U]$	**(2)** Controlled coeff. β (SE) $[R^2_C]$	**(3)** Identified set $[\beta,\ \beta^*(\min\{1, 1.3R^2_C\}, \delta = 1)]$	**(4)** δ for $\beta = 0$ given $R^2_{\max}$
Panel A: Composite NCS Index				
IU-Age 5 × Target	-0.217	-0.226	$[-0.226, -0.326]$	5.827
(SE)	(0.087)	(0.081)		
$[R^2]$	[0.059]	[0.106]		
Panel B: Agreeableness				
IU-Age 5 × Target	-0.317	-0.314	$[-0.314, -0.246]$	1.723
(SE)	(0.119)	(0.130)		
$[R^2]$	[0.065]	[0.095]		
Panel C: Conscientiousness				
IU-Age 5 × Target	-0.141	-0.162	$[-0.162, -0.035]$	1.143
(SE)	(0.146)	(0.153)		
$[R^2]$	[0.046]	[0.066]		
Panel D: Extraversion				
IU-Age 5 × Target	-0.027	-0.073	$[-0.073, -0.109]$	8.629
(SE)	(0.162)	(0.172)		
$[R^2]$	[0.051]	[0.076]		
Panel E: Neuroticism				
IU-Age 5 × Target	-0.283	-0.140	$[-0.140, -0.223]$	16.841
(SE)	(0.169)	(0.147)		
$[R^2]$	[0.099]	[0.235]		
Panel F: Openness				
IU-Age 5 × Target	-0.313	-0.437	$[-0.437, -0.822]$	-2.254
(SE)	(0.183)	(0.192)		
$[R^2]$	[0.067]	[0.098]		

Notes: Each panel reports results for one outcome. Column (1) is the uncontrolled coefficient (fixed effects only); column (2) adds the full control set. The identified set spans the controlled coefficient and the bias-adjusted coefficient beta*(delta=1), with R-squared max set to 1.3 times the controlled R-squared (Oster 2019). Column (4) reports the delta that would shrink the coefficient to zero; delta greater than one (or negative) indicates the estimate is unlikely to be driven by unobservable (Oster 2019; Graham et al. 2017). Estimates are survey weighted with standard errors clustered on state.

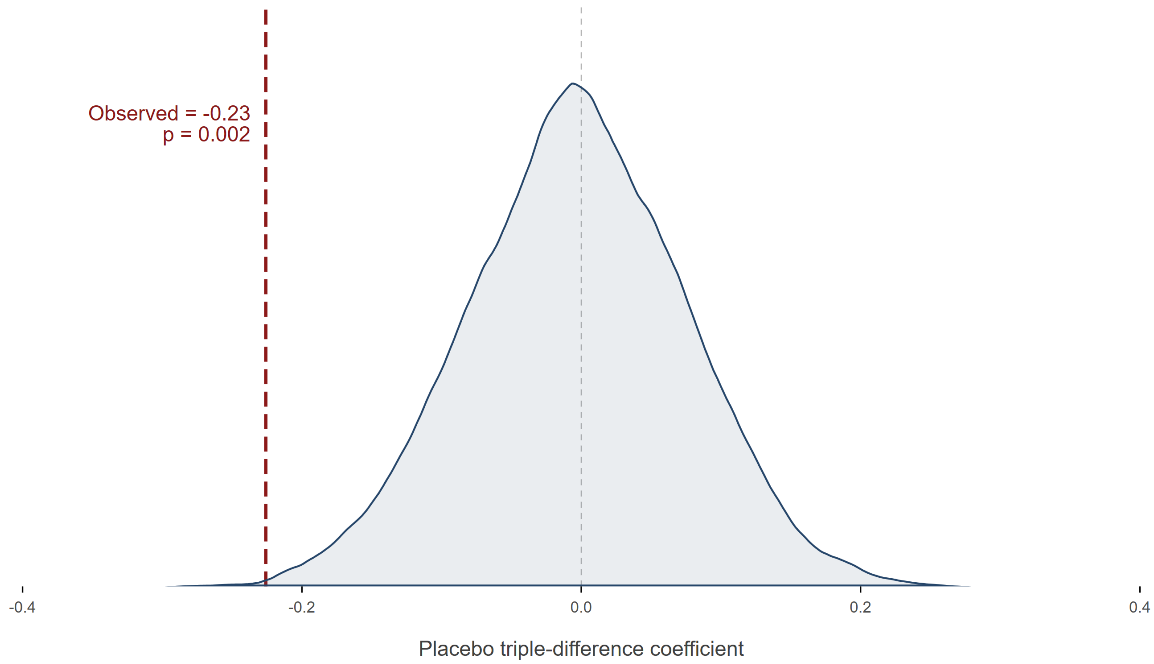

Fig. 3: Randomization Inference: Placebo Target Assignment and the Baseline DDD Estimate
Notes: This figure presents a non-parametric randomization-inference (permutation) test of the composite DDD estimate. Target status is randomly reassigned across individuals and the placebo DDD coefficient recomputed 1,000 times, building the distribution of estimates expected if treatment eligibility carried no signal. The shaded curve is the placebo distribution and the vertical line the observed effect; the placebo distribution centers near zero and the observed estimate lies in its extreme tail, indicating the result is very unlikely to arise by chance. Being distribution-free, the test complements the state-clustered standard errors.

Appendix

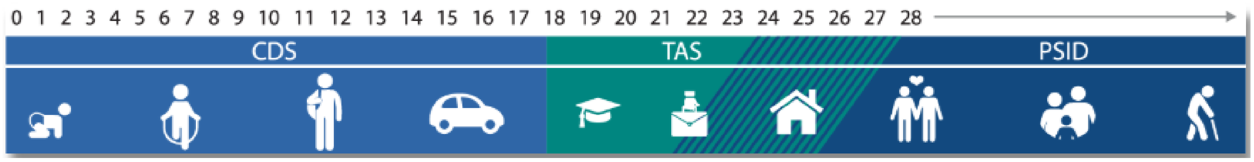

Fig. A1: Time of how CDS, TAS, and PSID map onto children's development.
Notes: This figure illustrates the longitudinal structure of the PSID-CDS-TAS data used in this study. The CDS sampled children born between 1983 and 1997 at ages 0 to 12 at baseline (wave I) and followed them through age 17. Upon turning 18, these same individuals transitioned into the TAS, which tracks them biennially through approximately age 28.

Table A1: State Welfare Reform Implementation Dates

State	Waiver Date	TANF Date	State	Waiver Date	TANF Date
Alabama		11/15/1996	Montana	2/1/1996	2/1/1997
Alaska		7/1/1997	Nebraska	10/1/1995	12/1/1996
Arizona	11/1/1995	10/1/1996	Nevada		12/3/1996
Arkansas	7/1/1994	7/1/1997	New Hampshire		10/1/1996
California	12/1/1992	1/1/1998	New Jersey	10/1/1992	7/1/1997
Colorado		7/1/1997	New Mexico		7/1/1997
Connecticut	1/1/1996	10/1/1996	New York		11/1/1997
Delaware	10/1/1995	3/10/1997	North Carolina	7/1/1996	1/1/1997
District of Columbia		3/1/1997	North Dakota		7/1/1997
Florida		10/1/1996	Ohio	7/1/1996	10/1/1996
Georgia	1/1/1994	1/1/1997	Oklahoma		10/1/1996
Idaho		7/1/1997	Oregon	2/1/1993	10/1/1996
Illinois	11/23/1993	7/1/1997	Pennsylvania		3/3/1997
Indiana	5/1/1995	10/1/1996	Rhode Island		5/1/1997
Iowa	10/1/1993	1/1/1997	South Carolina		10/12/1996
Kansas		10/1/1996	South Dakota	6/1/1994	12/1/1996
Kentucky		10/18/1996	Tennessee	9/1/1996	10/1/1996
Louisiana		1/1/1997	Texas	6/1/1996	11/5/1996
Maine		11/1/1996	Utah	1/1/1993	10/1/1996
Maryland	3/1/1996	12/9/1996	Virginia	7/1/1995	2/1/1997
Massachusetts	11/1/1995	9/30/1996	Washington	1/1/1996	1/10/1997
Michigan	10/1/1992	9/30/1996	West Virginia	2/1/1996	1/11/1997
Minnesota		7/1/1997	Wisconsin	1/1/1996	9/1/1997
Mississippi	10/1/1995	7/1/1997	Wyoming		1/1/1997
Missouri	6/1/1995	12/1/1996			

Notes: This table shows the exact dates adopted either a waiver or the TANF program across the United States. Source: U.S. Department of Health and Human Services.

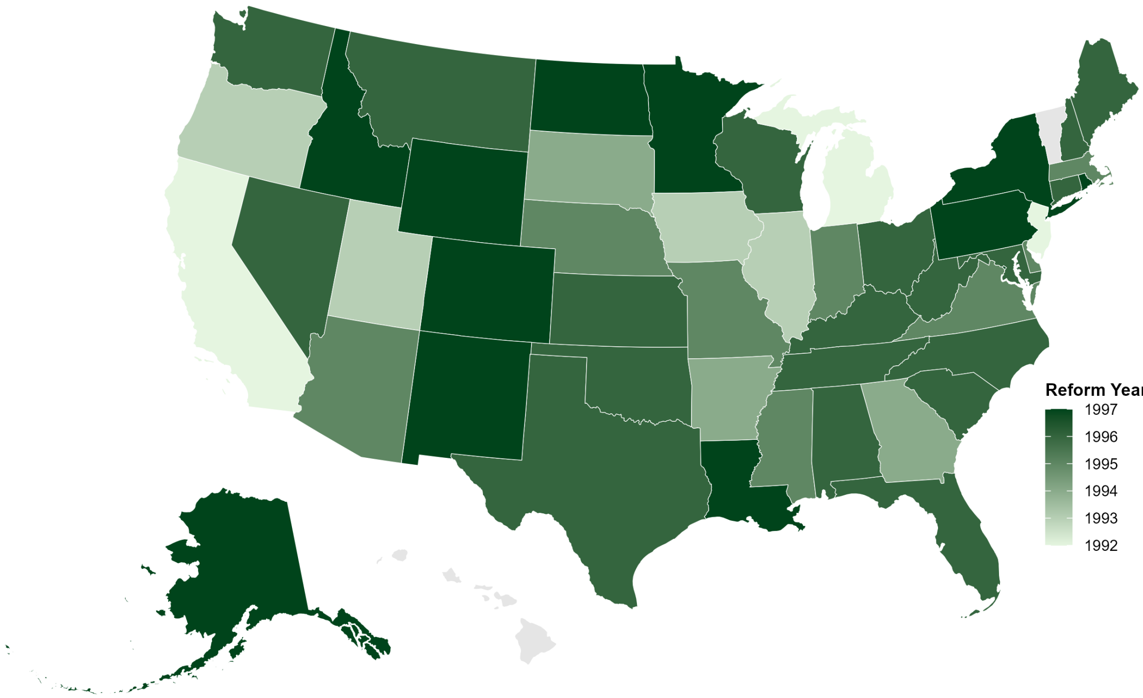

Fig. A2: Welfare Reform Implementation Timing by State
Notes: This figure shows the timing of state-level welfare reform implementation across the United States. Source: U.S. Department of Health and Human Services.

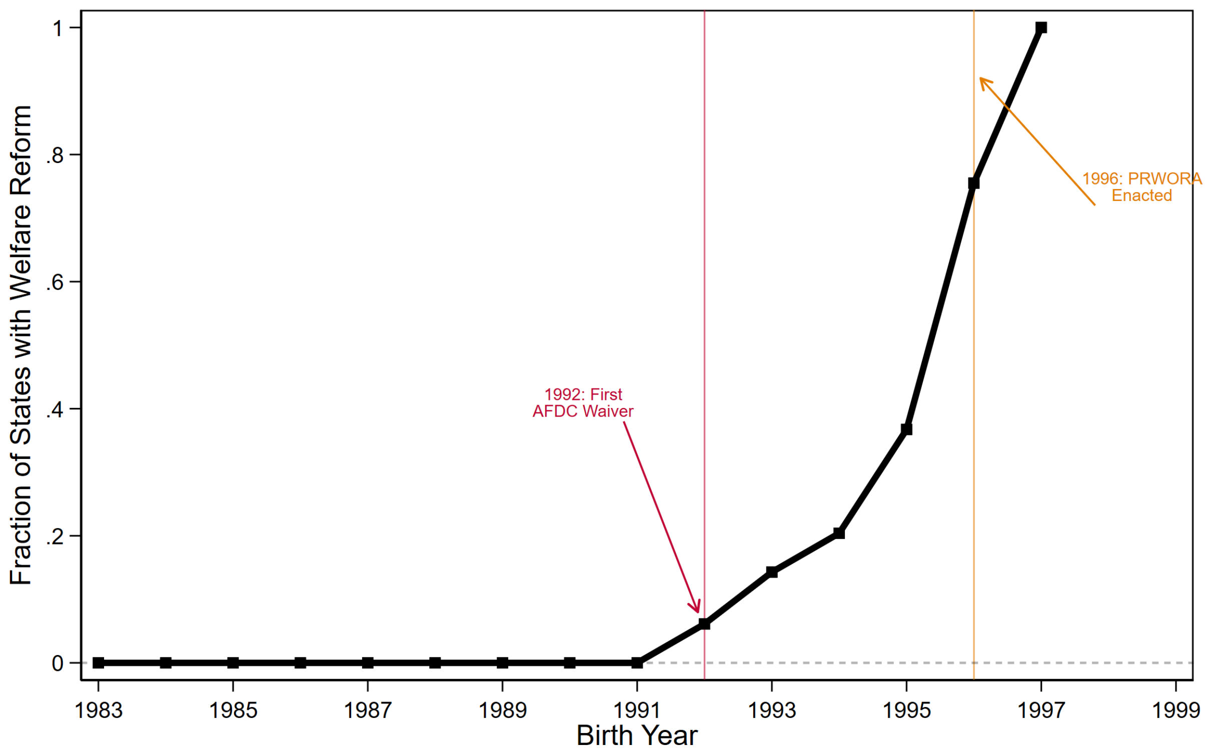

Fig. A3: Staggered Rollout of State Welfare Reform
Notes: This figure shows the staggered rollout of 1990s welfare reform across states, plotting the fraction of states that had implemented reform against child birth year. A state is coded as reformed for a given cohort if the child was born on or after the state's reform adoption date (the earlier of its waiver or TANF implementation). The vertical lines mark the first AFDC waiver (1992) and the enactment of PRWORA (1996).

Table A2: Balance Table

	Control Mean	Target Mean	p-value	SMD
Young Adult Controls				
Age at Survey	23.410	23.572	0.303	0.060
Female	0.477	0.542	0.125	0.130
Nonwhite	0.201	0.626	0.000	0.957
Grandparent Present	0.020	0.050	0.241	0.169
Maternal Controls				
Mother's Age	34.437	32.526	0.067	0.234
< High School	0.098	0.448	0.000	0.854
High School	0.269	0.552	0.000	0.601
Some College	0.543	0.000	0.000	1.542
Pre-Reform State Level Characteristics				
Unemployment Rate	6.260	6.615	0.041	0.264
ln(TANF Benefit)	12.476	12.519	0.749	0.043
ln(SNAP Benefit)	13.088	13.188	0.283	0.123
Poverty Rate	13.310	14.629	0.023	0.364
Minimum Wage	3.759	3.744	0.770	0.028
EITC Rate	0.010	0.019	0.423	0.126

Notes: This table reports weighted summary statistics for the target and control groups. Means are computed using TAS survey weights. The p-values are from two-sample t-tests of equality in means across groups. Standardized mean differences (SMDs) are calculated as the absolute difference in group means divided by the pooled standard deviation, where the pooled standard deviation is defined as the square root of the average of the two group-specific variances. Following common practice, an SMD below 0.10 indicates good covariate balance. Sample weights applied.

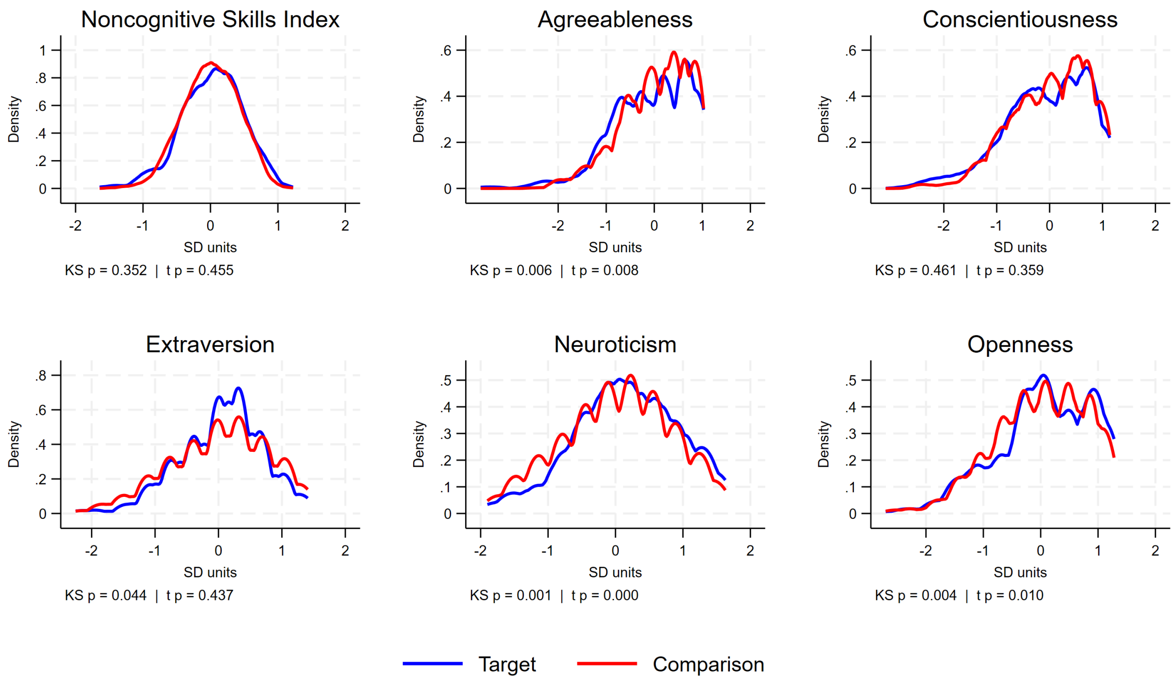

Fig. A4: Distribution of Noncognitive Skills: Target vs. Comparison Children
Notes: This figure plots survey-weighted kernel density estimates of the standardized composite NCS index and each Big Five subscale, comparing target children (born to single mothers with at most a high school education or less) with the comparison group (single mothers with more than high school education or from two-parent household). Distributions are evaluated on a common grid. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of distributions and from a two-sample t-test of equal means between the two groups.

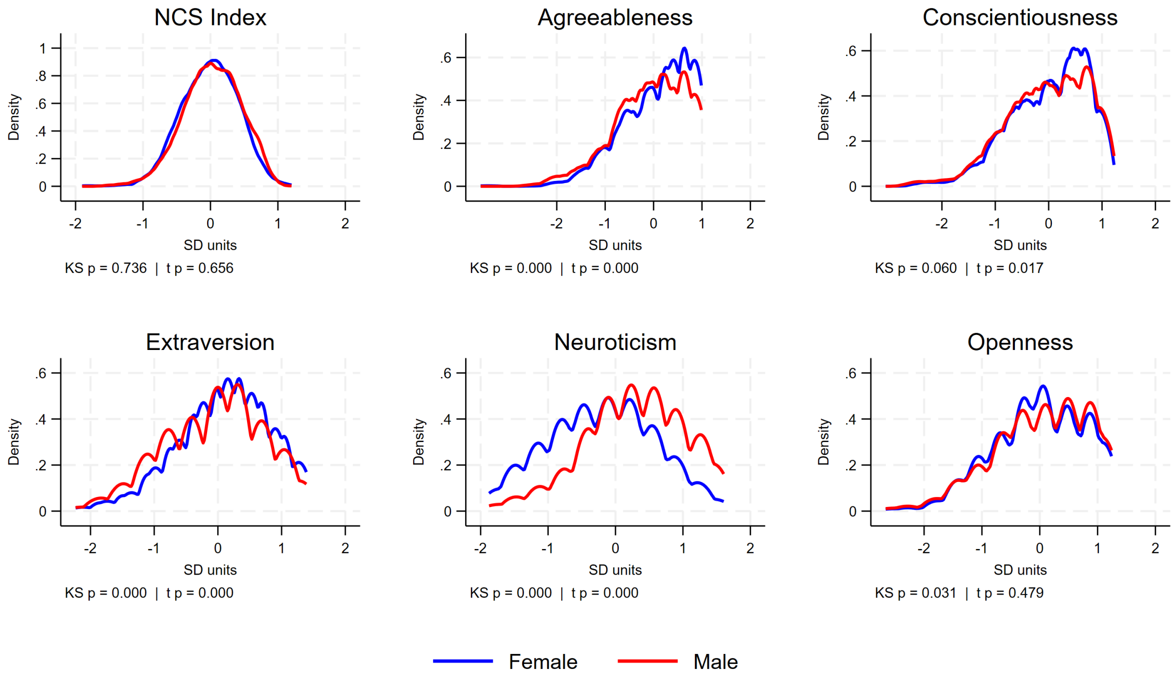

Fig. A5: Distribution of Noncognitive Skills, by Child Gender
Notes: This figure plots survey-weighted kernel density estimates of the standardized composite NCS index and each Big Five subscale, comparing female and male young adults. Distributions are evaluated on a common grid. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of distributions and from a two-sample t-test of equal means between the two groups.

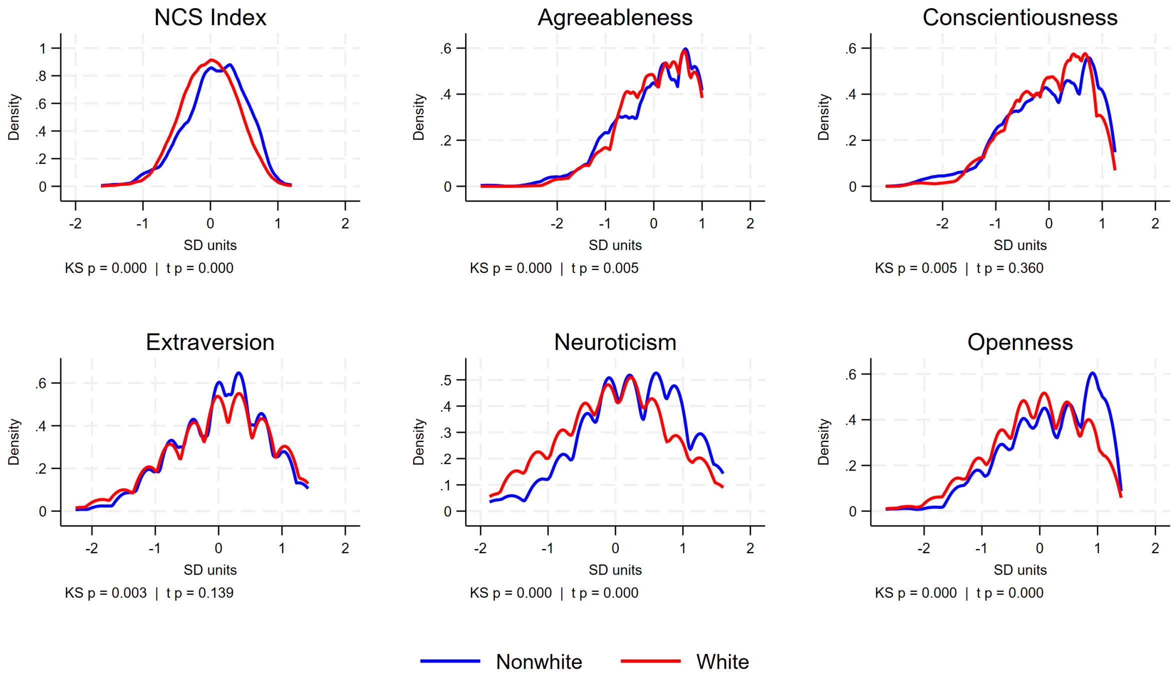

Fig. A6: Distribution of Noncognitive Skills, by Child Race
Notes: This figure plots survey-weighted kernel density estimates of the standardized composite NCS index and each Big Five subscale, comparing Nonwhite and White children. Distributions are evaluated on a common grid. Each panel reports the p-value from a Kolmogorov-Smirnov test of equality of distributions and from a two-sample t-test of equal means between the two groups.

Table A3: Reduced Form: Reform Exposure and Young Adult Labor Outcomes

	(1) Earnings (PPML)	(2) Employment (LPM)
Exposure IU-Age 5	0.288*	0.032
	(0.160)	(0.047)
IU-Age 5 × Target	-0.176	0.053
	(0.352)	(0.112)
Observations	2830	2833

Notes: This table reports reduced-form DDD estimates of the effect of early-life welfare-reform exposure on adult labor outcomes. Column (1) models earnings using Poisson pseudo-maximum-likelihood (PPML); column (2) models employment using a linear probability model. "IU-Age 5" measures the fraction of the conception-to-age-5 window occurring under post-reform rules; "Target" identifies children born to mothers with at most a high school education or less, and IU-Age 5 × Target is the DDD estimate. Both specifications include maternal, young adult, and pre-reform state economic controls, together with birth-cohort, survey-year, target, and state fixed effects. Estimates are survey weighted. Standard errors, clustered at the state level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

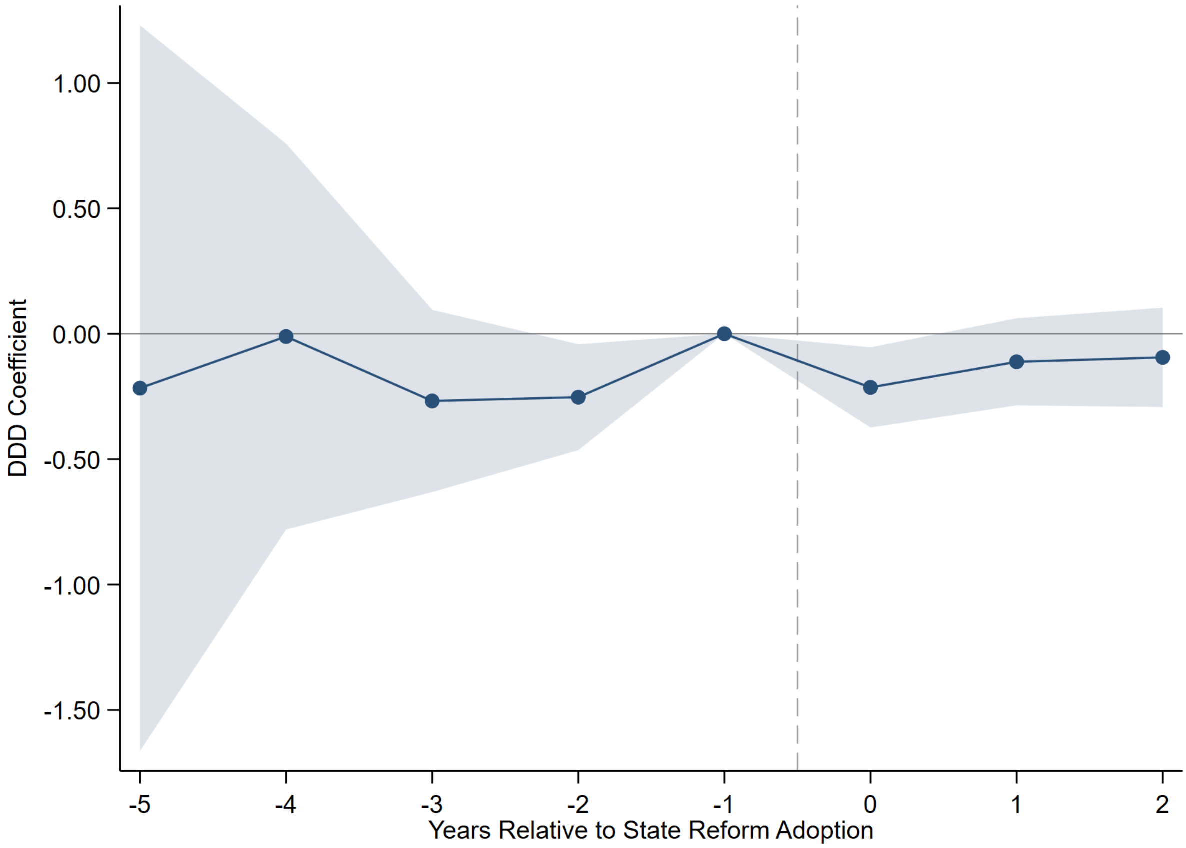

Fig. A7: Event-Study Estimates by Year Relative to State Reform Adoption (Stacked DDD)
Notes: This figure plots the dynamic DDD coefficients from a stacked event-study design for the composite NCS index. Because no state remains untreated within the sample window, each adoption cohort is matched to not-yet-treated states (those adopting later), following Cengiz et al. (2019), and the sub-experiments are stacked with cohort-specific fixed effects. Coefficients are measured relative to the omitted reference year ($t = -1$), with post-adoption years at $t \geq 2$ pooled. The joint test of the pre-adoption coefficients yields $p = 0.275$, supporting parallel trends. Shaded bands are 90% confidence intervals from standard errors clustered at the state level; estimates are survey weighted.

Table A4: Survey Items for the Big Five Personality Scales

Personality Factor	Definition	Survey Items
Agreeableness	The tendency to act in a cooperative, unselfish manner.	Sometimes rude to others *(R)* Forgiving Considerate and kind to others
Conscientiousness	The tendency to be organized, responsible, and hardworking.	Thorough worker Somewhat lazy *(R)* Effective and efficient Completes tasks
Extraversion	An orientation of one's interests and energies toward the outer world of people and things, characterized by positive affect and sociability.	Communicative, talkative Outgoing, sociable Reserved *(R)*
Neuroticism / Emotional Stability	Emotional stability reflects predictability and consistency in emotional reactions, with an absence of rapid mood changes; neuroticism reflects a chronic level of emotional instability and proneness to distress.	Worrier *(R)* Nervous *(R)* Relaxed, able to deal with stress
Openness to Experience	The tendency to be open to new aesthetic, cultural, or intellectual experiences.	Original, comes up with new ideas Values artistic, aesthetic experience Imaginative

Notes: Items are drawn from the PSID Transition into Adulthood self-report personality battery. Definitions follow the American Psychological Association Dictionary of Psychology. Items marked (R) are reverse-coded prior to scale construction so that higher values denote more of the named trait.

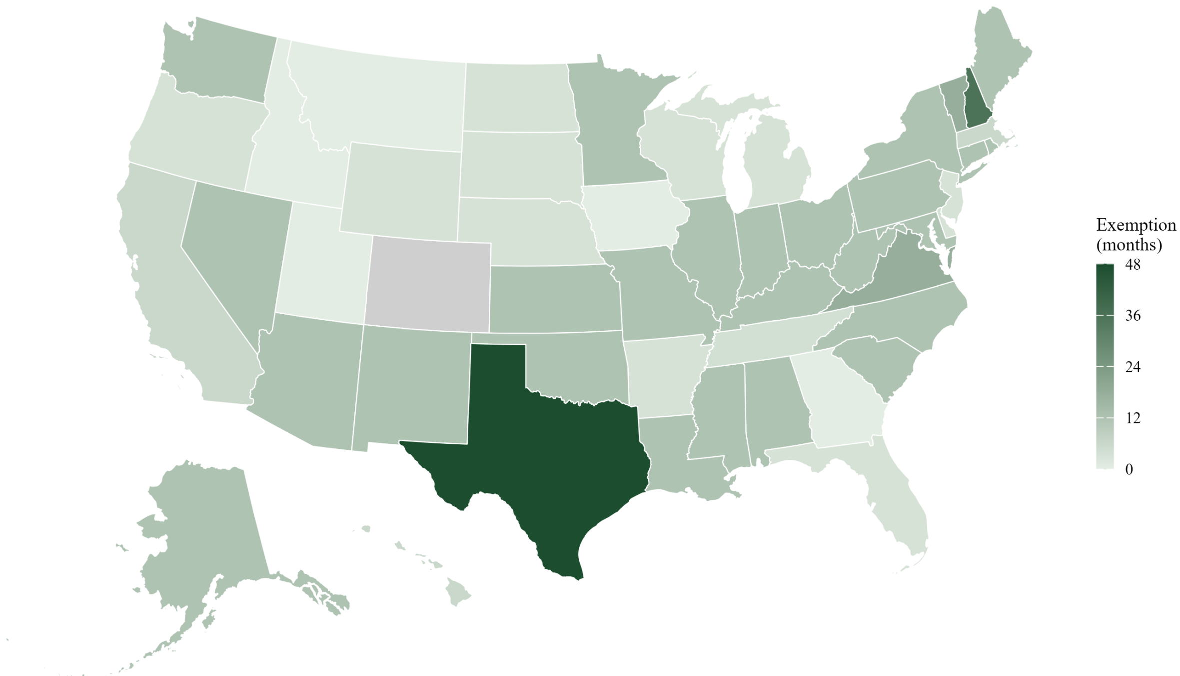

Fig. A8: Age-of-Youngest-Child Work Exemption under TANF by State

Table A5: Age-of-Youngest-Child Work Exemption (AYCE) under TANF, by State

Exemption threshold (age of youngest child at which the parent must participate in work activity), converted to months.

State	TANF Exemption Threshold	Months	State	TANF Exemption Threshold	Months
Alabama	1 year	12	Nebraska	3 months	3
Alaska	1 year	12	Nevada	1 year	12
Arizona	1 year	12	New Hampshire	3 years	36
Arkansas	3 months	3	New Jersey	12 weeks	3
California	6 months	6	New Mexico	1 year	12
Colorado	county	missing	New York	1 year	12
Connecticut	1 year	12	North Carolina	1 year	12
Delaware	13 weeks	3	North Dakota	3 months	3
DC	1 year	12	Ohio	1 year	12
Florida	3 months	3	Oklahoma	1 year	12
Georgia	No	0	Oregon	90 days	3
Hawaii	6 months	6	Pennsylvania	1 year	12
Idaho	No	0	Rhode Island	1 year	12
Illinois	1 year	12	South Carolina	1 year	12
Indiana	1 year	12	South Dakota	12 weeks	3
Iowa	No	0	Tennessee	4 months	4
Kansas	1 year	12	Texas	4 years	48
Kentucky	1 year	12	Utah	No	0
Louisiana	1 year	12	Vermont	18 months	18
Maine	1 year	12	Virginia	18 months	18
Maryland	1 year	12	Washington	1 year	12
Massachusetts	6 months	6	West Virginia	1 year	12
Michigan	3 months	3	Wisconsin	12 weeks	3
Minnesota	1 year	12	Wyoming	3 months	3
Mississippi	1 year	12			
Missouri	1 year	12			
Montana	No	0			

Notes: Age-of-youngest-child work exemption under TANF, from ASPE Table W-2.a (Setting the Baseline: A Report on State Welfare Waivers). The threshold is the age of the youngest child at or below which a parent is exempt from work requirements; higher values denote more protective policies. Thresholds are converted to months (1 year = 12; 13 weeks, 12 weeks, and 90 days = 3). States coded No exemption are set to 0 (least generous). Colorado sets exemptions at the county level and is coded missing. This threshold is the exempt-months moderator used in the AYCE heterogeneity analysis.

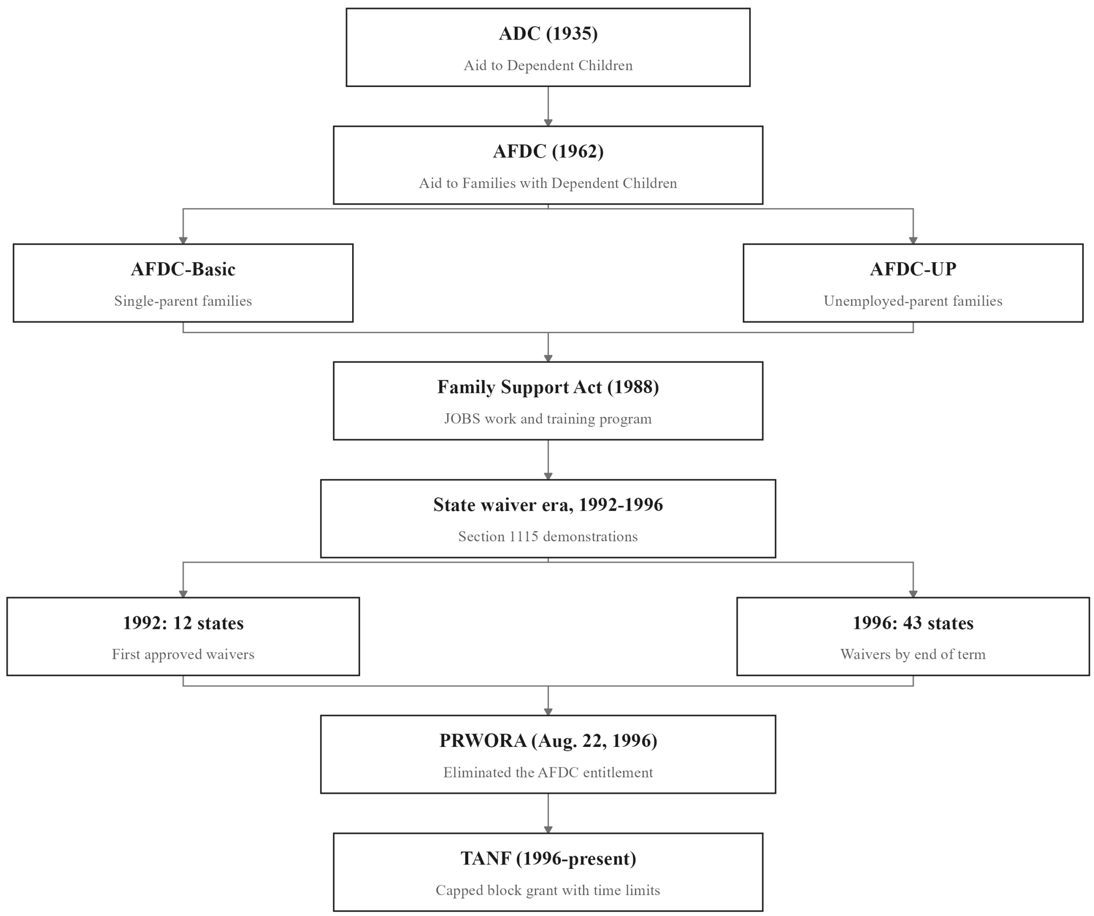

Fig. A9: Welfare Reform Genealogy Plot

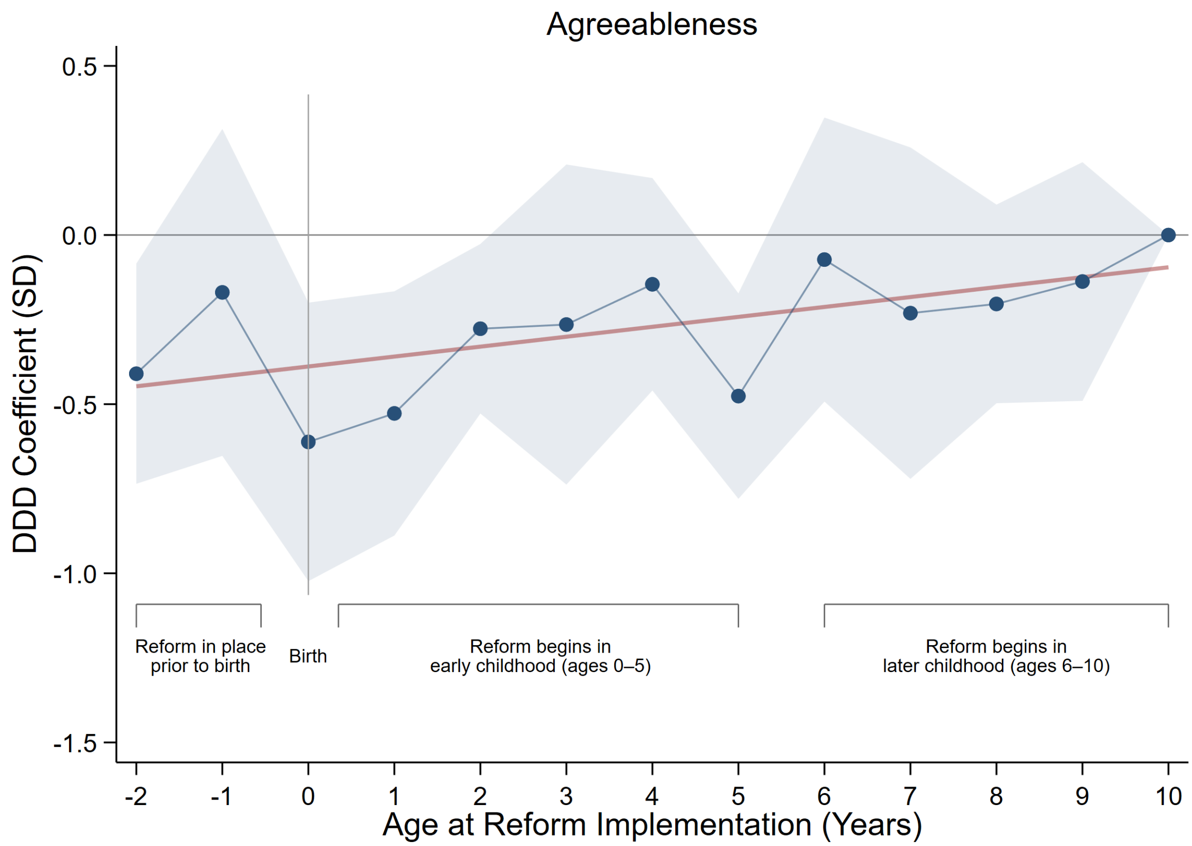

Fig. A10: Effect of Welfare Reform by Age at Exposure: Agreeableness

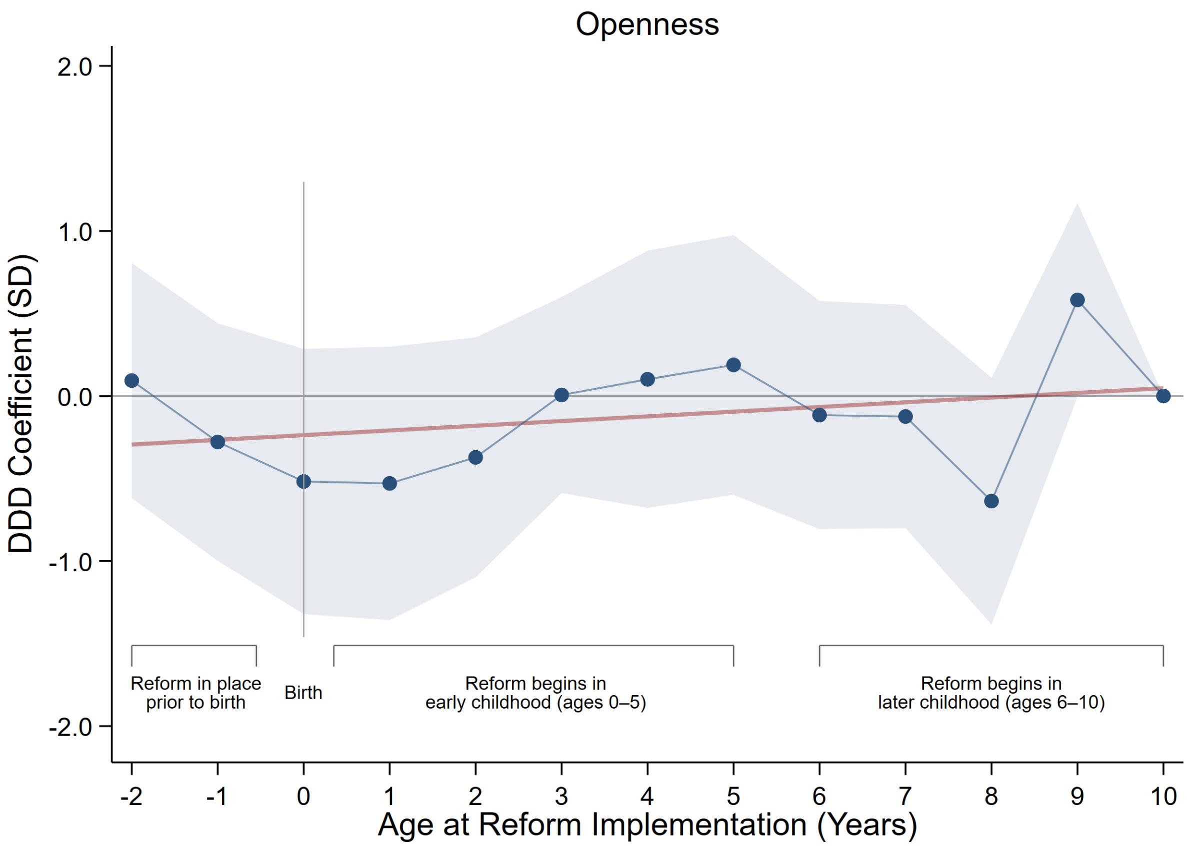

Fig. A11: Effect of Welfare Reform by Age at Exposure: Openness

Technical Appendix

Appendix A. Construction of the NCS Index

Let i index individuals and j index the noncognitive items derived from the Big Five battery. All items are reported on a 1 to 4 scale. For some items a higher raw score corresponds to a worse trait (for example, "rude," "lazy," "nervous"); for others a higher score corresponds to a better trait (for example, "kind," "forgiving").

A.1. Reverse-coding of negatively framed items

Let r_{ij} denote the raw response of individual i to item j. Let N be the set of negatively framed items (rude, lazy, reserved, worries, nervous) and P the remaining positively framed items. I first map all items to a common direction in which higher values indicate more favorable NCS,

$$x_{ij} = \begin{cases} 5 - r_{ij} & \text{if } j \in N, \\ r_{ij} & \text{if } j \in P. \end{cases} \tag{A1}$$

By construction x_{ij} remains on the 1 to 4 scale, with larger values uniformly interpretable as more favorable traits.

A.2. Control-group standardization (Kling-style z-scores)

Let $D_i = 1$ if child i is in the target group (a mother with at most a high school education) and $D_i = 0$ if in the comparison group. Following Kling et al. (2007), I standardize each item using the comparison group only, so that effects are expressed in standard-deviation units relative to the untreated distribution. For each item j, I compute the comparison-group mean and standard deviation,

$$\mu_j = E[x_{ij} \mid D_i = 0], \qquad \sigma_j = SD[x_{ij} \mid D_i = 0], \tag{A2}$$

and define the standardized score for individual i on item j as

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \tag{A3}$$

so that $z_{ij} = 1$ means child i lies one comparison-group standard deviation above the comparison-group mean on trait j.

A.3. Overall noncognitive index

The overall NCS index is the simple average of the 15 standardized items. Let J_i denote the set of non-missing items for individual i. The index is

$$NCS_i = \frac{1}{|J_i|} \sum_{j \in J_i} z_{ij} \tag{A4}$$

If all items are observed then $|J_i| = 15$; otherwise the mean is taken over the available items. By construction, among comparison-group children the index has mean approximately zero and variance close to one, so every treatment coefficient reads as a standard-deviation change in overall NCS relative to the comparison group.

Appendix B. Construction of the Welfare-Reform Exposure Variable

The treatment variable measures the share of a child's earliest developmental window spent under post-reform welfare rules. This appendix documents its construction for the primary in-utero to age-five window.

B.1. Timing anchors

For each child I observe the birth year and birth month. I date conception to nine months before birth, so the conception month is the birth month minus nine (rolling into the prior calendar year when the difference is non-positive), with the conception year decremented accordingly. The exposure window runs from conception through the fifth birthday, a span of 69 months (9 months in utero plus 60 months of life), and all exposure shares use 69 months as the denominator.

B.2. Assigning reform dates to states

Each child is assigned to a state, the state of residence in the 1997 CDS. To that state I attach two policy dates recorded to the month, the implementation date of its AFDC Section 1115 waiver where one was granted, and the implementation date of its TANF program under PRWORA. Reform timing is the earlier of the two, so a state is treated from the first date it departed from the pre-reform AFDC regime.

B.3. Months of exposure under each regime

For the waiver regime I count the window-months falling on or after the waiver date. Let the waiver be implemented in month m_w of year y_w, and let conception occur in month m_c of year y_c. The waiver-exposed months are

$$E_i^w = 12(y_w - y_c) + (m_w - m_c), \tag{B1}$$

counting how much of the 69-month window lies at or after the waiver date. This quantity is bottom-coded at 0 (children whose window closed before the waiver) and top-coded at

69 (children conceived entirely under the waiver). Because a state's waiver was superseded once TANF took effect, waiver exposure is additionally capped at the waiver's own duration, the number of months between the waiver and TANF dates, so months after TANF are not double-counted. TANF-exposed months E_i^T are computed by the identical formula using the TANF date, bottom-coded at 0 and top-coded at 69.

B.4. Interview-date correction

Some children had not reached age five at the 1997 CDS interview, so their window was not yet complete. For these children I cap exposure at the share of the window actually elapsed by the interview date, subtracting any excess so that exposure never exceeds the portion of the window a child had lived. Children already past age five completed the full window and are unaffected.

B.5. Total exposure and the treatment interaction

Each regime's exposure is converted to a share by dividing by 69, and total early-life exposure is the sum of the waiver and TANF shares, capped at one,

$$RE_i^{\text{IU-5}} = \min\left\{\frac{E_i^w + E_i^T}{69},\ 1\right\}. \tag{B2}$$

A value of 0 denotes a child whose entire window predated any reform in their state, and a value of 1 a child who spent the whole window under post-reform rules. The continuous-dose DDD treatment is the interaction of this exposure share with target status,

$$RE_i^{\text{IU-5}} \times T_i \tag{B3}$$

which is the coefficient of interest throughout the paper.

Appendix C. Robustness Methods

This appendix sets out the three methods used to probe the robustness of the baseline triple-difference estimate, the stacked estimator that guards against negative weighting under staggered adoption, the permutation test that provides distribution-free inference, and the Oster (2019) procedure that bounds the scope for omitted-variable bias.

C.1. Stacked Triple-Difference Estimator

Under staggered adoption with heterogeneous effects, the two-way fixed-effects estimator can form forbidden comparisons in which already-treated units serve as controls for later-treated ones, delivering a weighted average of treatment effects in which some weights are negative (Goodman-Bacon, 2021; de Chaisemartin and d'Haultfoeuille, 2020). Because every state in my sample adopts reform between 1992 and 1996, no never-treated group is available. The stacked design (Cengiz et al., 2019; Wing et al., 2024) circumvents this by building a set of clean sub-experiments, each with its own not-yet-treated controls, and estimating the baseline specification on the pooled stack.

Step 1. Define adoption cohorts. Let c index the calendar years in which states adopt reform, $c \in \{1992, \ldots, 1996\}$. Each adoption year defines one sub-experiment, indexed by e.

Step 2. Assign treated and control units. For sub-experiment e tied to adoption year c, the treated group comprises children born in states adopting in year c. The control group comprises children in not-yet-treated states, those adopting strictly after c. States adopting before c are excluded, since they are already treated and would reintroduce forbidden comparisons.

Step 3. Trim to a common event window. Within each sub-experiment I retain birth cohorts in a symmetric window around the adoption year, from five cohorts before to four after, so every sub-experiment contributes a comparable range of relative cohorts and thin boundary cells are excluded.

Step 4. Stack the sub-experiments. The trimmed sub-experiments are appended into a single dataset. A child near a cohort boundary may appear in more than one sub-experiment under a different reference each time; this is a feature of the design, not double counting, because identification proceeds within each e.

Step 5. Estimate with sub-experiment-specific fixed effects. On the stacked sample I estimate the same triple difference as in the main text, with every fixed effect interacted with the sub-experiment index e,

$$NCS_{isce} = \delta_1 RE_{isc}^{\text{IU-5}} + \delta_2 T_i + \delta_3 \left(RE_{isc}^{\text{IU-5}} \times T_i\right) + \beta X_{isce} + \alpha_{se} + \gamma_{ce} + \eta_{te} + \theta_{Te} + \varepsilon_{isce} \quad \text{(C1)}$$

where α_{se}, γ_{ce}, η_{te}, and θ_{Te} are state, birth-cohort, survey-year, and target fixed effects specific to sub-experiment e. The estimator is identical to the baseline; only the sample and the indexing of the fixed effects change. Because these fixed effects confine every contrast to within a single sub-experiment, treated cohorts are compared only against not-yet-treated states, and the forbidden comparisons that generate negative weights cannot arise. The coefficient δ_3 recovers the triple-difference effect, and standard errors are clustered on state.

C.2. Permutation Inference

Treatment varies at the state level and the number of clusters is moderate, so the asymptotic approximation underlying clustered standard errors may be unreliable. I complement conventional inference with a non-parametric randomization test (Ferrara et al., 2012) that asks how often an effect as large as the observed one would arise if treatment eligibility carried no signal.

Step 1. Record the observed estimate. Estimate the baseline specification and store the coefficient on the interaction, $\hat{\beta}$.

Step 2. Construct a placebo treatment. For replication r, randomly reassign target status across individuals, drawing a placebo indicator $T_i^{(r)}$ that preserves the share of treated units but severs any true link to family background. Form the placebo interaction $RE_{isc}^{\text{IU-5}} \times T_i^{(r)}$.

Step 3. Re-estimate and store the placebo coefficient. Re-estimate the baseline specification with the placebo interaction in place of the true one, and store the coefficient $\hat{\beta}^{(r)}$.

Step 4. Repeat to build the null distribution. Repeat Steps 2 and 3 for $R = 1{,}000$ replications, yielding an empirical distribution $\{\hat{\beta}^{(r)}\}$ that approximates the distribution of the estimate under the null of no effect. Because target status is assigned at random, this distribution centers on zero.

Step 5. Compute the randomization p-value. The two-sided permutation p-value is the share of placebo estimates at least as large in magnitude as the observed coefficient,

$$p = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\left(|\hat{\beta}^{(r)}| \geq |\hat{\beta}|\right) \tag{C2}$$

Because it is built entirely from the data through random relabeling, the test is distribution-free and makes no appeal to large-sample theory. An observed estimate lying in the extreme tail of the placebo distribution, with a correspondingly small p, indicates the effect is very unlikely to have arisen by chance.

C.3. Bounding Omitted-Variable Bias (Oster, 2019)

Selection on unobservables cannot be ruled out by controls alone. Oster (2019) formalizes the intuition that if adding observed controls moves the coefficient little while raising the R-squared, then unobservables of comparable explanatory power would also move it little. The procedure uses the movement of the coefficient and the R-squared between an uncontrolled and a controlled regression to bound the bias.

Step 1. Estimate the uncontrolled and controlled regressions. Let $\tilde{\beta}$ and $\tilde{R}$ be the coefficient and R-squared from a regression with only the fixed effects, and $\dot{\beta}$ and $\dot{R}$ the coefficient and R-squared from the controlled regression that adds the full set of covariates.

Step 2. Set the maximum R-squared. Specify $R_{\max}$, the R-squared a regression on both observed and unobserved determinants would attain. Following common practice I set $R_{\max} = \min\{1.3\,\dot{R},\ 1\}$, allowing unobservables to explain up to thirty percent more variation than the observed controls.

Step 3. Compute the bias-adjusted coefficient. Under equal selection on observables and unobservables ($\delta = 1$), the bias-adjusted coefficient is approximately

$$\beta^* \approx \dot{\beta} - (\tilde{\beta} - \dot{\beta})\frac{R_{\max} - \dot{R}}{\dot{R} - \tilde{R}} \tag{C3}$$

The identified set for the true effect is the interval bounded by the controlled coefficient $\dot{\beta}$ and the bias-adjusted β^*. An identified set that excludes zero indicates the effect survives proportional selection on unobservables.

Step 4. Compute the critical δ. Alternatively, fix the target at $\beta = 0$ and solve for the δ required to attribute the entire effect to unobservables. This δ measures how large selection on unobservables would have to be, relative to selection on the observed controls, to drive the coefficient to zero. A value greater than one means unobservables would need to be more important than the full set of observed controls, an implausibly strong requirement; a negative suggests that omitted-variable bias, if proportional to selection on observables, would strengthen rather than weaken the estimated effect.

Step 5. Interpret the bounds. I report both objects for each outcome, the identified set from equation (C3) and the critical δ for $\beta = 0$. Coefficient stability across the addition of controls, an identified set excluding zero, and a large or negative critical δ together indicate that the estimate is unlikely to be an artifact of unobserved confounding.